

Logic Manager Control Functions

LM09-600

**Implementation
Logic Manager**

Logic Manager Control Functions

LM09-600

R682

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About This Publication

This is a reference manual for process engineers, control system engineers, and application engineers who design and implement data acquisition and control strategies for **TotalPlant** Solution (TPS) systems with Local Control Networks.

The user should be familiar with the system control functions described in *System Control Functions* in the *Implementation/Startup & Reconfiguration - 2* binder before using this publication.

Detailed descriptions of the parameters mentioned in this publication can be found in the *Logic Manager Parameter Reference Dictionary* in this binder, and further background information relative to Logic Manager implementation can be found in *Logic Manager Implementation Guidelines* located in the *Implementation/Logic Manager* binder.

This publication supports **TotalPlant** Solution (TPS) system network Release 682. TPS is the evolution of TDC 3000^X.

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1. INTRODUCTION

This section contains an overview of the Logic Manager, its subsystems and ladder logic devices, and also provides references to other publications that are useful or necessary during implementation.

1.1 OVERVIEW

The Logic Manager (LM) is a process-connected device that resides on the **TotalPlant** Solution (TPS) Universal Control Network as a peer to the Process Manager. The LM consists of a standard Honeywell 620-35 Logic Controller (LC) that contains a two-slot option module known as the Logic Manager Module (LMM), and is mounted in one or more standard TPS cabinets. See Figure 1-1.

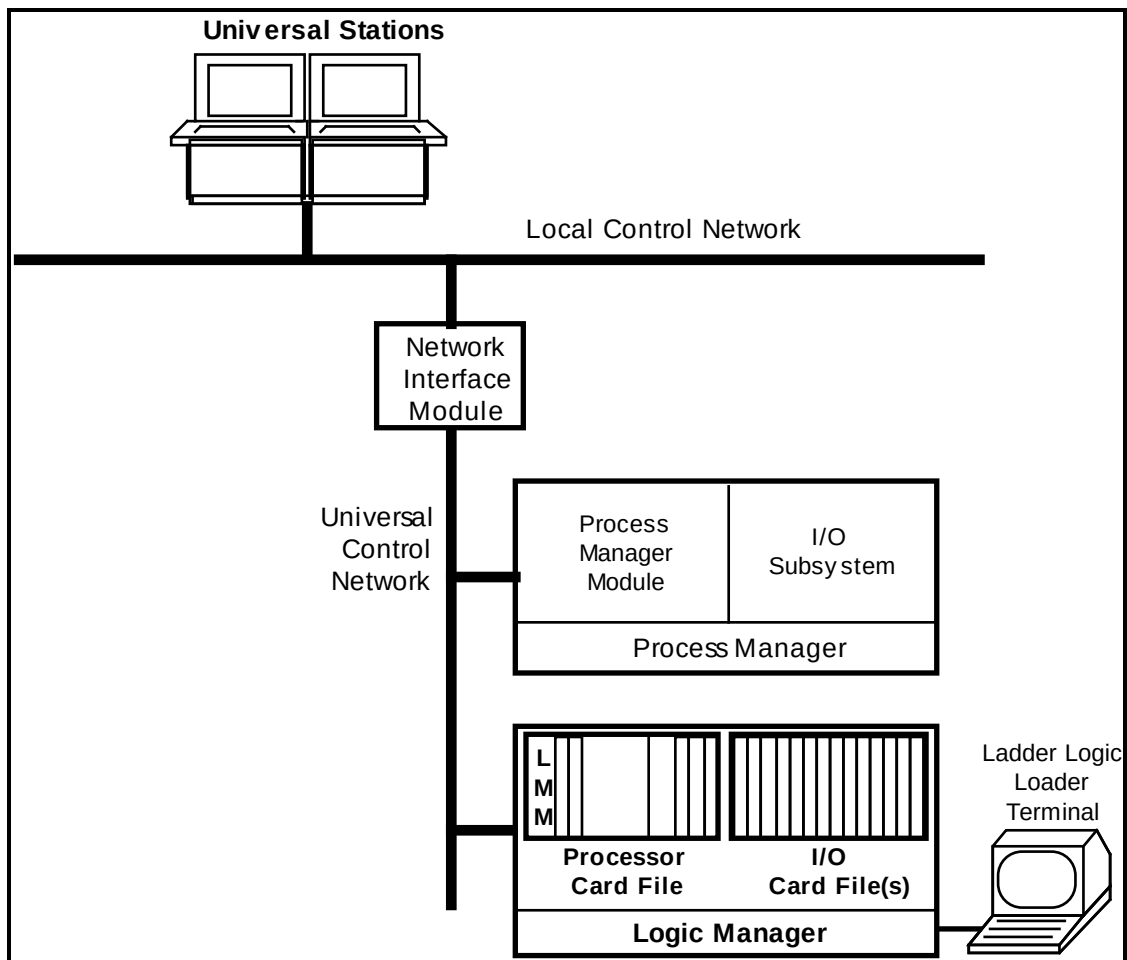


Figure 1-1 — Logic Manager as a Component of TPS

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As shown in Figure 1-1, the Honeywell 620 Logic Controller uses two basic types of racks to house the numerous kinds of available plug-in modules:

- Processor Rack
- I/O Rack

In addition, a programming device to enter the ladder logic is required. These are described in the following sections.

1.2 LM PROCESSOR SUBSYSTEM

The Processor card file serves as the central “brain” for the system. Figure 1-2 shows the module layout of the non-redundant processor card file. The standard configuration contains the following modules:

- Logic Manager Module — 2-slot
- Honeywell 620-35 Processor (also called the “LM Processor”) — consists of 5 sub-modules:
 - Memory Module
 - Register Module
 - System Control Module
 - Processor Module
 - I/O Control Module
- Power Supply Module — 2-slot
- Serial Link Module(s)
- Parallel Link Driver Module

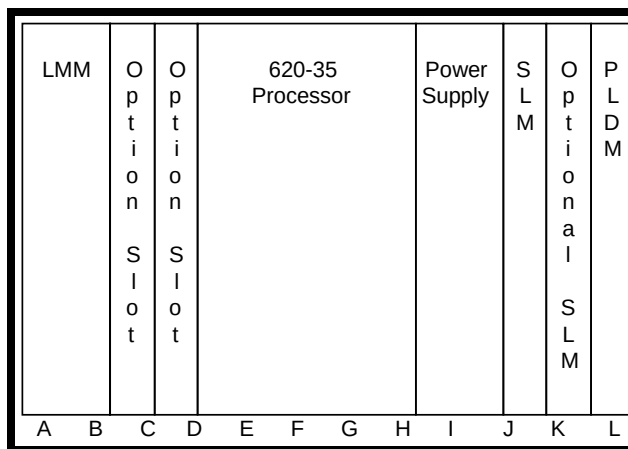


Figure 1-2 — Modules in Nonredundant Processor Card File

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1.2.1 Logic Manager Module (LMM)

The Logic Manager Module serves as the interface between the Honeywell 620-35 Logic Controller and the Universal Control Network. It exchanges data between its own tables and the tables of the Logic Controller. It collects and processes information from the 620 Logic Controller and ladder logic program; converts the information to TPS data types; and performs engineering unit conversions, alarm handling, annunciation, diagnostic status reporting, peer-to-peer communications, and other UCN functions. It processes information and converts it to the appropriate formats for passage to the Logic Controller. Its UCN communication channel is redundant.

1.2.2 Memory Module (MM)

The Memory Module holds the ladder logic program. Model No. 620-0024, with a capacity of 24k words is provided with the Logic Manager. Additional memory modules are not available for the Logic Manager.

1.2.3 Register Module (RM)

The Register Module contains the Logic Controller's I/O data tables. Much of the information exchanged between the Logic Controller and the Logic Manager Module is transferred from or into these RM tables. See subsection *Register Module Data Tables* of this manual for a description of table layout. Register Module model no. 620-0056 (4k x 4k) is provided with the Logic Manager.

1.2.4 System Control Module (SCM)

The System Control Module (model no. 620-0054), a functional extension of the Processor Module, coordinates the interaction of all the modules in the processor. The SCM also contains a single-bit processor that solves all relay logic in the control program.

1.2.5 Processor Module (PM)

The Processor Module (model no. 620-0080) executes the program stored in the memory module and handles arithmetic computation and data movement instructions.

1.2.6 I/O Control Module (IOCM)

The Input/Output Control Module coordinates communications between the Logic Manager processor and the 621 I/O systems, and formats the data flowing between the I/O system and the processor. It also monitors individual I/O module fault diagnostics. The IOCM works in conjunction with the Parallel Link Driver Module and/or Serial Link Module to control data flow to me /O. Model no. 620- 0085, with a capacity for 2048 I/O points (2040 points with serial I/O) is provided with the Logic Manager.

1.2.7 Power Supply Module (PSM)

The Power Supply Module provides power to operate the processor. A battery compartment contains a lithium battery, which provides backup power to the memory and register modules. Both 115/230 Vac and 24 Vdc models are available.

1.2.8 Serial Link Module (SLM)

The Serial Link Module provides the interface to the remote serial I/O subsystem (described below). One SLM is provided with every Logic Manager, and a reserved slot in the processor rack can accommodate the second one.

1.2.9 Parallel Link Driver Module (PLDM)

The Parallel Link Driver Module works with the IOCM to control I/O communications. The PLDM also controls system status by a processor mode keyswitch on the front panel, and it determines I/O response to system faults and various operating parameters. The PLDM helps control LM system I/O whether it be parallel I/O or serial I/O.

The LM's PLDM keyswitch has four positions for selecting processor mode of operation. The positions are labeled: PROGRAM, DISABLE, RUN/PROG, and RUN.

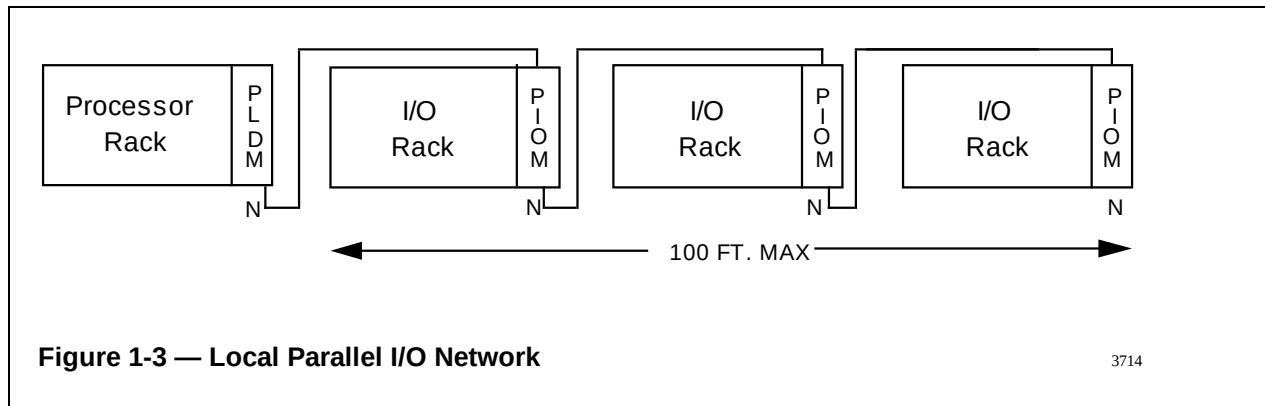
1.3 LM I/O SUBSYSTEM

I/O racks, like the processor racks, have 14 module slots. The I/O racks accommodate 12 single-width I/O modules, a power supply, and an interface module. The interface module can be one of two types—one for a local parallel I/O system and the other for a remote serial I/O system. The two types can be combined in a single Logic Manager.

1.3.1 Local Parallel I/O Subsystem

The Parallel I/O Subsystem is designed for those applications in which long distances between the I/O racks and the processor are not required. Access time is extremely fast. The first I/O rack of the local parallel I/O system is connected to the Parallel Link Driver Module (PLDM) in the Logic Manager processor rack. The Parallel I/O Module (PIOM) is installed in slot N of each I/O rack (Figure 1-3). The LM supports 2048 I/O points in a parallel I/O configuration.

The 620-9937 PIOM allows any combination of 8-, 16- or 32-point modules in an I/O rack, providing the total number of I/O points does not exceed 2048. A 100-foot parallel transmission length is permitted with each Logic Manager system.



1.3.2 Remote Serial I/O Subsystem

The remote serial I/O configuration allows I/O racks to be mounted close to the machine or the process being controlled at distances up-to-10,000 feet from the LM processor. This eliminates long multiple wire runs from the field devices to the area where the processor is located. Serial I/O update is independent of the Logic Manager processor and therefore scan time is unaffected by serial I/O.

The 621-9939 Serial Link Module (SLM) resides in slot L and optionally in slot M in the Logic Manager processor rack. The SLM communicates with the 621-9938R or 621-9940 Serial I/O Module (SIOM) located in each remote rack (slot N).

Each SLM operates two channels and up-to-two SLMs may be installed within one Logic Manager processor rack. Each channel supports up-to-16 I/O racks or a maximum of 2040 I/O points. Each I/O slot of a serial I/O rack can be configured to accommodate 0-, 8-, 16-, or 32-point I/O modules.

Figures 1-4 and 1-5 illustrate remote serial I/O system multi-drop installation patterns.

1.3.3 I/O Modules

The I/O subsystem offers many different types of input and output modules. These modules contain varying numbers of points and voltage types designed for compatibility with the various kinds of field devices and power sources to which I/O modules are connected:

- Digital Input
- Analog Input
- Digital Output
- Analog Output
- Special Function

Appendix A lists the I/O modules available for the Logic Manager. For detailed descriptions of each type, refer to the *620 I/O Specifications User Manual* (form 620-8995). Note that the user manual lists several modules that are not valid options for use with the Logic Manager.

I/O modules that have an “R” suffix may be inserted/removed from an I/O card file under power if the 621-9938R SIOM is used in that I/O card file.

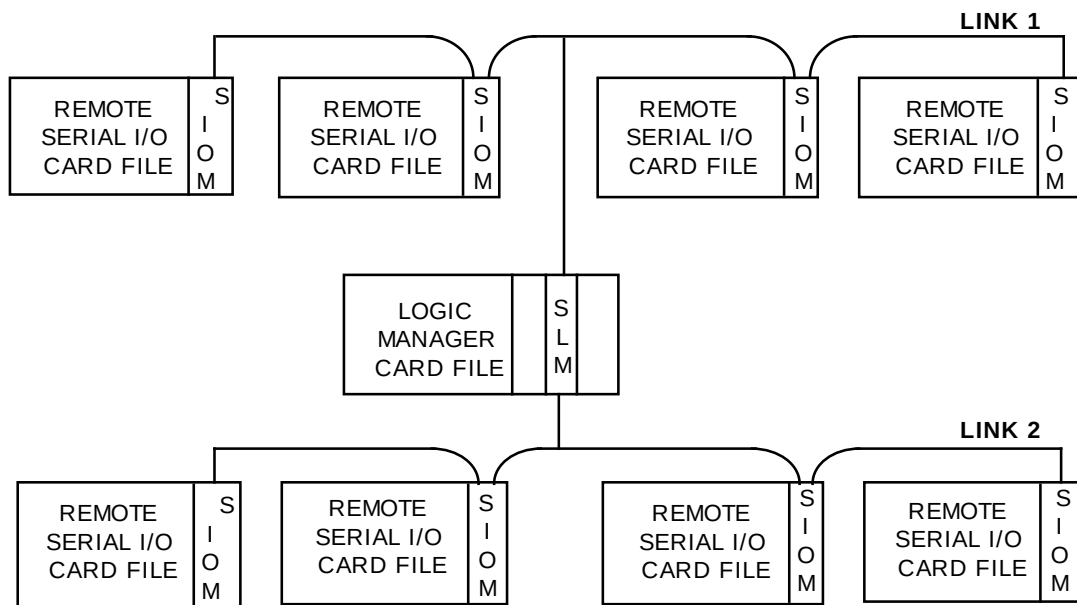


Figure 1-4 — Serial I/O Multidrop Network

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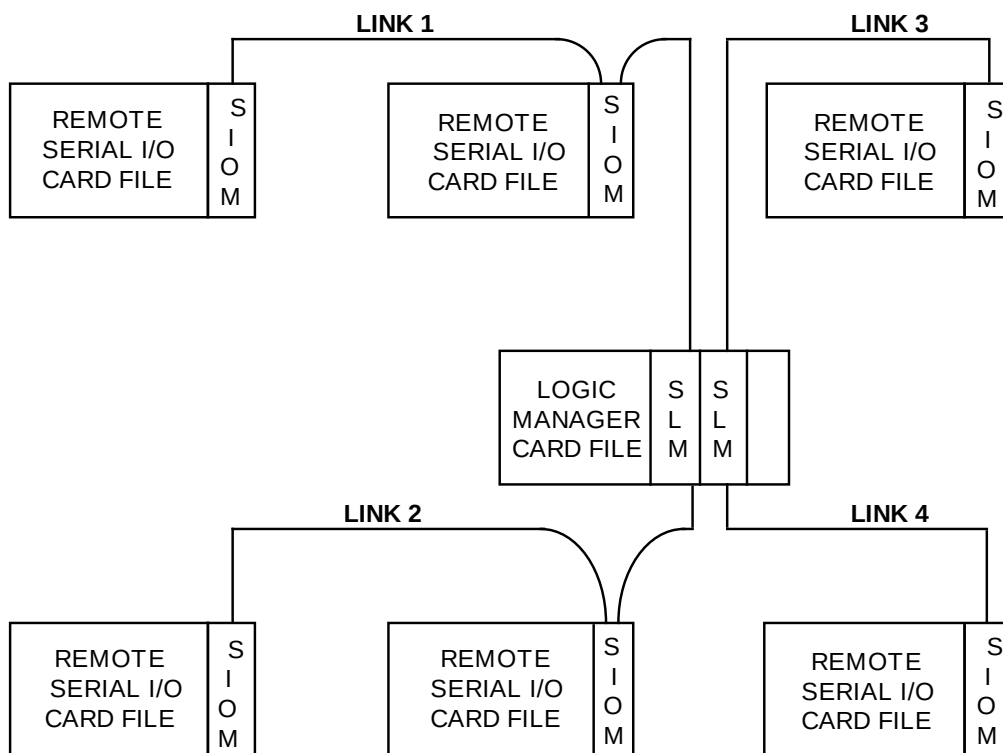


Figure 1-5 — Serial I/O Multidrop Network

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1.4 PROGRAMMING DEVICES AND LADDER LOGIC

The Logic Manager controls its local parallel or remote serial I/O by utilizing a powerful instruction set within the LM's processor module. The Logic Manager is programmed using ladder logic that is entered and loaded by the 623-6010/6020 MS-DOS Loader software loaded onto an IBM-compatible personal computer.

The 623-6010 MS-DOS Loader Software package includes program disks, an internal RS422 interface board, and special cable. The 623-6020 MS-DOS Loader Software package includes program disks, an external RS232/422 converter, and cables for use with an existing personal computer serial port. When installed, both packages transform almost any IBM-compatible personal computer into a Logic Manager programming device.

For more information on either programming product, please refer to the following Honeywell 620 User Manual:

623 MS-DOS Loader Model 623-60 User Manual (Form 632-8982)

For a list and brief description of program instructions, refer to Appendix B.

1.5 OPERATOR INTERFACE

The Universal Station standard displays associated with the Logic Manager and its data points are consistent with similar displays provided for the Process Manager and other TPS devices. Figure 1-6 shows Logic Manager ladder logic load and save targets as they appear on a standard UCN Status Display.

ENTER FILE NAME: 21 Sep 88 13:46:35 1

UCN CABLE STATUS: OK UCN 10 STATUS UCN CONTROL STATE: FULL
UCN AUTO CHECKPNT: ENABLE

01 NIM OK BACKUP	03 PM OK BACKUP	05 PM ALIVE ALIVE	07 LM OK BACKUP				

LOAD/SAVE RESTORE CONTROL STATES AUTO CHECKPT UCN CABLE STATUS RUN STATES SLOT SUMMARY DETAIL STATUS

PROGRAM LOAD RESTORE DATA SAVE DATA LMLADDER LOAD LMLADDER SAVE ENTER CANCEL

Figure 1-6 — UCN Status Display Showing LM Targets

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1.6 REFERENCES

The following publications contain information related to the control functions described in this publication:

- **General LM Implementation Details** — *Logic Manager Implementation Guidelines*, in the *Implementation/Logic Manager* binder.
- **Definitions of the LM Parameters** — *Logic Manager Parameter Reference Dictionary*, in the *Implementation/Logic Manager* binder.
- **Logic Controller MS-DOS Loader and Ladder Logic Instructions** — *623 MS-DOS Loader Model 623-60 User Manual* (Form 623-8982).
- **Logic Controller I/O Module Specifications** — *620 I/O Specifications User Manual* (Form 620-8995).
- **Control Functions in the System** — *System Control Functions*, in the *Implementation/Startup & Reconfiguration - 2* binder.
- **Control Functions in the Process manager** — *Process Manager Control Functions and Algorithms*, in the *Implementation/Process Manager - 2* binder.
- **Control Functions in the Advanced Process Manager** — *Advanced Process Manager Control Functions and Algorithms*, in the *Implementation/Advanced Process Manager - 2* binder.
- **Control Functions in the Application Module** — *Application Module Control Functions*, in the *Implementation/Application Module - 1* binder.

2. LM FUNCTIONS

This section provides a brief explanation of the various functions within the Logic Manager. For further implementation details, see Section 32 of the Engineer's Reference Manual, located in the Implementation/Startup and Reconfiguration - 2 binder.

2.1 LOGIC CONTROLLER OPERATION

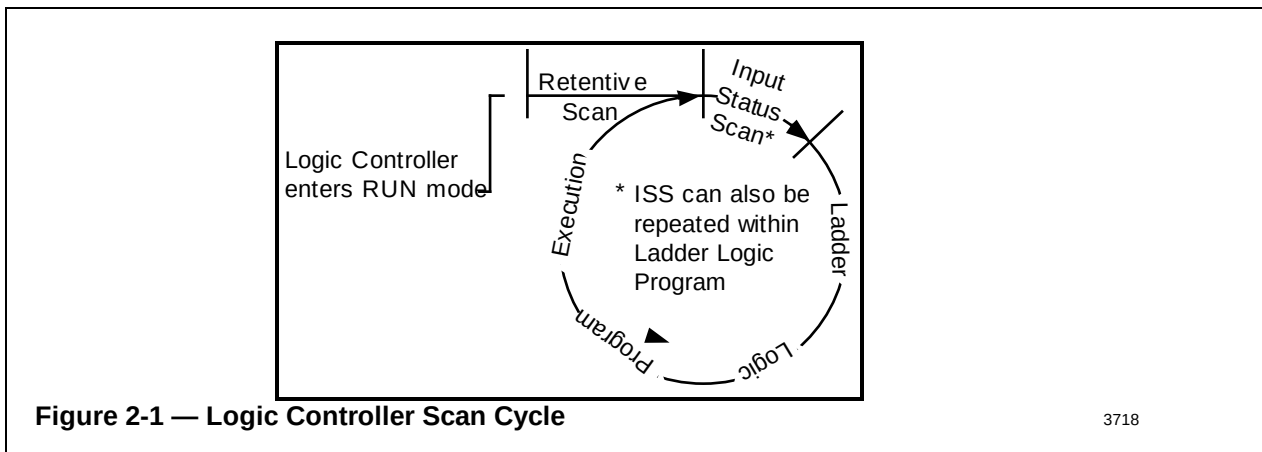
The Logic Manager features four primary modes of operation: RUN, PROGRAM, RUN/PROGRAM, and DISABLE. There is also a sub-mode called SOFTWARE PROGRAM MODE, which can be initiated when in either the RUN/PROGRAM or DISABLE primary mode. Control over the operating mode is exercised through the four-position keyswitch on the front panel of the Parallel Link Driver Module, although the mode can also be changed under certain conditions from the Loader/Terminal or Communications Interface Module. This is discussed further in the following paragraphs.

2.1.1 Run Mode

The system is in the RUN mode when the front panel keyswitch is in the RUN or RUN/PROGRAM position. The RUN mode is the main control mode for the Logic Manager. Figure 2-1 illustrates the normal Logic Controller scan cycle. It is important to recognize that the Logic Manager Module operates on its own scan cycle, which is asynchronous to the LC cycle.

The system executes a retentive scan when it first enters the RUN mode. During the retentive scan all non-retentive outputs from 0 to 4095 are turned OFF. The retentive outputs retain the state they were in during the last scan executed before being removed from the RUN mode.

After the retentive scan is complete, the user program scan begins by verifying that an Input Status Scan (ISS) instruction is located in the first memory location of the user program.



During the ISS time window, some other functions are performed in addition to the collection of input status from the I/O subsystem. This includes diagnostics and option card communication.

The Logic Manager then does a program memory scan by reading the second location in the Memory Module and continues through the user ladder logic program until it encounters a *Return to Beginning of Program* or *End of Memory* instruction. Either of these instructions causes the scanning sequence to repeat, beginning with a new ISS. In this instance, a *Return to Beginning of Program* is an optional instruction programmed by the user. The *End of Memory* instruction is automatically deposited in the Memory Module by the device that loaded the program into memory.

2.1.2 Program Mode

When the Logic Manager is in the PROGRAM mode, a signal is transmitted to the I/O System, which allows individual I/O racks to be selected to freeze or clear outputs. Contacts may be forced if the Force Enable Switch (SW2 switch 4) on the PLDM is CLOSED/ON. Timer/counter data stored in the Register Module may be changed.

The system can be placed in the PROGRAM mode by the Loader/Terminal or some other remote programming device. If this is the case, the Software PROGRAM mode request must be removed from the Logic Manager, causing the system to return to the mode of operation specified by the position of the key switch.

2.1.3 Run/Program Mode

RUN/PROGRAM mode is used for making program changes. When the Logic Manager system is in the RUN/PROGRAM mode the system operates identically to the RUN mode with the added capability of program editing. The user's program is executed as it is in the RUN mode.

The RUN/PROGRAM permits the following functions: element status may be forced by the Loader/Terminal if the force enabled switch on the PLDM is closed, and preset and accumulated values of timer and counters may be changed by the Loader/Terminal if the data change enable switch on the PLDM is closed.

2.1.3.1 Augmented Run Mode Programming (ARMP)

The capability to make program changes while in the RUN/PROGRAM mode has been refined with the introduction of Augmented RUN Mode Programming (ARMP). This feature allows users to make control program additions and selections while in the RUN/PROGRAM mode with no adverse affect on system operation, other than temporarily increased scan time (20 msec maximum).

2.1.4 Disable Mode

The DISABLE mode can be entered by default as well as through the front panel keyswitch. DISABLE by default occurs if the system does not detect a selected mode.

When the system is in the DISABLE mode the Logic Manager scans through the user's program executing the instructions as it would for normal machine control operation. At the same time, the hardware on the Parallel Link Driver Module sends a power fail signal to the I/O system. Outputs are cleared or held in the last state in the DISABLE mode depending on the PIOM setting for each rack. The outstroke for updating the I/O is held disabled, preventing I/O output updating.

The Logic Controller can be placed in the Software PROGRAM mode while in DISABLE. The DISABLE mode forces the Logic Manager Module into the IDLE state.

2.1.5 Software Program (Sub) Mode

The system can be placed in the Software PROGRAM mode by the MS-DOS Loader or a Communications Interface Module. The Logic Manager must be in the RUN/PROGRAM or DISABLE mode, and the on-line programming function enabled according to SW2 switch 6 on the PLDM. The system enters the Software PROGRAM mode only after the scan being executed is completed.

When the Loader removes the Software PROGRAM mode request, the Logic Manager leaves the software PROGRAM mode and returns to the original mode, after the system successfully executes the retentive scan and self-diagnostics.

Software PROGRAM mode changes are made through the MS-DOS Loader mode change auxiliary menu. This function is particularly useful in the program debug stage for extensive changes. The user may monitor program execution, find a bug, change it, and execute the program from the keyboard.

2.2 LADDER LOGIC PROGRAM

NOTE

When programming the ladder logic through the MS-DOS loader, remember that coordination with Logic Manager data point configuration at the Universal Station is critical. Failure to insure compatibility can have an unintended impact on Logic Manager performance.

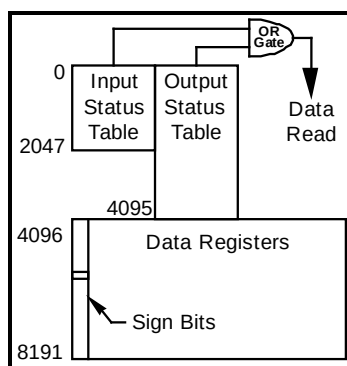


Figure 2-2 — RM Memory Map

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2.2.1 Register Module Data Tables

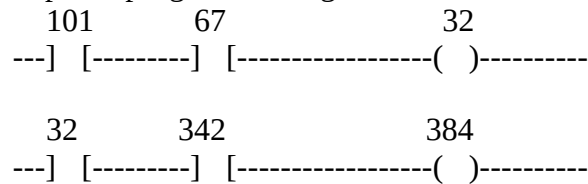
As each Input Status Scan is performed and followed by execution of the ladder logic, data is stored into and retrieved from sets of tables in the Register Module. Figure 2-2 shows how these tables are organized.

The status inputs from the Input Status Scan are stored in the Input Status Table, which is sized to accommodate a maximum of 2048 input points. As the logic program execution progresses, digital outputs are stored by the program in the Output Status Table, and are also sent to the I/O Control Module to update field outputs. Locations in the I/O table not connected to field I/O can be used by the program for other single-bit storage (internal coils). Numerical data is stored and retrieved by the program using the Data Registers (addresses 4096 through 8191).

Note the method employed for extracting data from the I/O Status Tables. When reads are made from internal addresses 0 through 2047, the result is obtained by performing a logical OR on the contents of the Input and Output Status Tables for the requested address.

2.2.2 Ladder Logic Program Example

Below is an example of programmed logic lines and how the program executes them.



1. The first instruction (-] [- 101) is brought from memory causing a logical OR to be performed on the bits stored at address 101 of the Input and Output Status Tables of the Register Module. Assuming that input 101 was ON at input status scan, a binary 1 would have been stored in the Input Status Table at address 101. All Output Status Table bit addresses are cleared to zeros during retentive scan, so the OR operation will result in a 1 (contact 101 in the logic line above CLOSED).
2. The next instruction (-] [- 67) is brought from memory and a logical OR performed on the Input and Output Status Tables at address 67. If input contact 67 is CLOSED, a binary 1 would have been stored in the Input Status Table at address 67, and the contact above will now be CLOSED.
3. The next instruction (-()- 32) is brought from memory. The Logic Manager recognizes that output 32 must be turned ON since both input contacts 101 and 67 are closed. The Logic Manager transmits a signal over the field I/O bus to energize output 32 and posts a binary 1 in address 32 of the Output Status Table.
4. The next instruction (-] [- 32) is brought from memory and a logical OR performed on the Input and Output Status Tables at address 32. Bit address 32 in the Input Status Table contains a 0, because no input module is in that position, but a binary 1 was just posted at address 32 in the Output Status Table as a result of the previous instruction. The result of the logical OR is a binary 1 (contact 32 closed).
5. The next instruction (-] [- 342) is brought from memory, and a logical OR performed on the Status Tables at address 342. If input contact 342 is OPEN, 0 was stored in the Input Status Table at address 342. Contact 342 above is therefore open.
6. The next instruction (-()- 384) is brought from memory. The Logic Manager recognizes that contact 32 is CLOSED, but contact 342 is OPEN, therefore output 384 must be de-energized. The Logic Manager transmits the signal via the field I/O bus to de-energize output address 384 and posts a 0 in bit address 384 of the Output Status Table.
7. The remainder of the instructions is executed until the End of Memory instruction is reached. The EOM returns the program execution sequence to the first memory word of the program (the Input Status Scan).

2.2.3 Ladder Logic Programming References

A list and brief description of program instructions that are valid for the Logic Manager are provided in Appendix B of this manual.

For detailed information on ladder logic programming of the Logic Manager, refer to the following *Logic Controller* user manual:

623 MS-DOS Loader Model 623-60 User Manual (Form 623-8982).

2.3 DATA POINT OVERVIEW

2.3.1 Summary of LM Data Point Types

Nine control-related data point types can be configured for the Logic Manager. Table 2-1 lists these points with a brief description of their function.

Table 2-1 — LM Data Point Types

Point Type	Description
Digital Input	Single-bit input from Logic Controller.
Digital Output	Single-bit output to Logic Controller.
Digital Composite	Multiple digital I/O point; can have 0-2 inputs, 0-3 outputs; typically used for motor control.
Analog Input	Real value from Logic Controller; may be raw process input from an analog input module or value supplied by the ladder logic program.
Analog Output	Real value output to a Logic Controller analog output module or to a data register for use by the ladder logic program.
Linkage	Also called <i>logic point</i> ; its function is to transfer data between UCN and Logic Controller connections. Supports peer-to-peer communications.
Flag	A 2-state point for storing a Boolean value.
Numeric	A point for storage of an integer value.
Timer	Provides timekeeping functions through interface to the Logic Controller timer capabilities.

2.3.2 Data Point Processing

Most data being transferred between the LMM and Logic Controller will be updated on a 0.5-second cycle. The only possible exceptions are analog input and analog output points, which can optionally be configured for either 0.5- or 1.0-second scan cycles.

NOTE

The LMM scan is not to be confused with the Logic Controller scan. The two processes are completely independent and run asynchronously.

Data Points are scanned in the following order:

1. Analog Input Points
2. Digital Input Points
3. Digital Composite Points
4. Logic Points
5. Timer Points
6. Flag Points (first 128 only)
7. Analog Output Points
8. Digital Output Points

This order of point processing cannot be configured or altered by the user.

NOTE

All UCN and Logic Manager input information is pre-fetched prior to their respective processing scan cycles. If this pre-fetch operation is not completed prior to normal scan, the cycle "slips" and data is retrieved on the next scan cycle. An excessive number of such scan cycle "slips" will result in a failure. All UCN and LM database output data is poststored for verification at a later scan cycle.

2.3.3 Data Point Mix

2.3.3.1 Point Capacity

As part of configuring the LM, the user chooses how many of each point type is to be implemented. Each point type has an absolute maximum limit per Logic Manager system as shown in Table 2-2. The LM points must be configured while the Logic Manager Module (LMM) is in the IDLE state.

Table 2-2 — Data Point Capacity

Type of Point	Maximum Quantity Per Point Type	Notes
Digital Input	1866	
Digital Output	1950	
Digital Composite	304	
Analog Input	127 254	0.5 second LMM scan rate 1 second LMM scan rate
Analog Output	482 965	0.5 second LMM scan rate 1 second LMM scan rate
Linkage	14	
Flag	1024	
Numeric	1024	
Timer	700	

2.3.3.2 Determining User Point Mix

Typically, the user configures a mix of different types of points. To assure that scanning of all configured points is achieved within the scheduled scan cycle, a per-point weighting system has been established.

Each LMM data point type has been assigned a unique weight, or *point processing unit* value. The user must multiply the intended number of each type of data point in a system by its point processing unit value to determine the total processing units that result from the desired point mix. The sum of the point processing unit totals of all LM Data Points cannot exceed 2800, which is the maximum predetermined Logic Manager Module processing unit total. Table 2-3 represents a user worksheet for making this point-mix calculation.

This page may be reproduced for use as a configuration worksheet.

Table 2-3 — Point Mix Worksheet

Point Type	Maximum Allowable Point Count	Processing Unit Value Per Point	X	Number or Points Desired	=	Total Point Processing Units
Digital Input	1866	1.3	X	_____	=	_____
Digital Output	1950	0.7	X	_____	=	_____
Digital Composite	304	9.2	X	_____	=	_____
Analog Input 1/2-second scan	127	22	X	_____	=	_____
Analog Input 1-second scan	254	11	X	_____	=	_____
Analog Output 1/2-second scan	482	5.8	X	_____	=	_____
Analog Output 1-second scan	965	2.9	X	_____	=	_____
Linkage (Logic)	14	200	X	_____	=	_____
Timer	700	4	X	_____	=	_____
POINT PROCESSING TOTAL _____ Note: Must be 2800 or less to be valid						

FLAG and NUMERIC points use a fixed amount of processing overhead and therefore are not required to be calculated into the Point Mix determination. The user can configure as many as 1,024 Flag points and as many as 1,024 Numeric points.

Table 2-4 is an example of an acceptable point mix that falls within the Logic Manager Module point processing unit limit of 2800.

Table 2-4 — Point Mix Example

Point Type	Maximum Allowable Point Count	Processing Unit Value Per Point	X	Number or Points Desired	=	Total Point Processing Units
Digital Input	1866	1.3	X	550	=	715
Digital Output	1950	0.7	X	410	=	287
Digital Composite	304	9.2	X	55	=	506
Analog Input 1/2-second scan	127	22	X	——	=	——
Analog Input 1-second scan	254	11	X	30	=	330
Analog Output 1/2-second scan	482	5.8	X	——	=	——
Analog Output 1-second scan	965	2.9	X	20	=	58
Linkage (Logic)	14	200	X	4	=	800
Timer	700	4	X	26	=	104
POINT PROCESSING TOTAL						2800
NOTE: This Point Mix is within the 2800 limit and therefore is valid.						

2.3.3.3 Backup and Restore box parameters

With TPN R682, you can take a backup of the box parameters to a file and later use the file for restoring the values.

You can take a backup or restore the box parameters through DEB and DEBEC. In DEB you can perform a backup or restore operation using the BOX PARAM BACKUP RSTR target. In DEBEC you can use the commands “BPP” and “BPR” for backup and restore operations respectively.

The backup and restore option is available for the following box parameter type:

- NUMERICS
- FLAGS

LM must be in run state (IDLE, OK) to perform a backup/restore. Multiple users can save backups in different locations. You can take a backup of the same parameter from different workstations at the same time. Similar to other DEB functionalities, file manager errors are reported, if the same file is selected by more than one user at the same time.

NOTE

Do not store the default values to Numerics. If the default option is selected, an unsuccessful list is created for the parameters that are not in the file.

Section 7 of the Data Entity Builder Manual explains how to perform the backup and restore operation for the box parameters.

2.3.3.4 UCN Device Resource Utilization Display

With R680, the number of available/consumed PUs can be viewed through the **UCN Device Resource Utilization** display. The display can be accessed through the target UCNDEVUT in the PERFMENU or LM CONFIG in the Control Configuration display of an LM.

The schematic displays the following resource utilization statistics for each slot type:

- Total number of PUs consumed by the LM node.
- Maximum number of PUs available for an LM node.
- Number of free units in PUs for an LM node.
- % loaded.

The free units in PUs are indicated in different colors as follows:

- If the free unit is greater than 30%, the value is backlit in green.
- If the free unit is between 20% and 30%, the value is backlit in yellow.
- If the free unit is less than 20%, the value is backlit in red.

In addition to the Resource Utilization statistics, the schematic displays the following:

- Number of points configured per slot type
- Point mix of the selected LM node
- Available PUs

Section 21 of the Process Operations Manual explains how to view the UCN Device Resource Utilization Display. Section 10 of the Engineer's Reference Manual explains the details of the displays.

2.3.4 Execution States

A loaded data point (with the exception of Flag and Numeric data points, which are scanned even when the LMM is in the IDLE mode) is in one of two execution states—active or inactive.

When in the active state, the point is being processed by the Logic Manager.

When in the inactive state the point is not being processed. A new or changed point will initially be placed in the inactive state when being loaded into the LM database. A point must be placed in this state before changes to its configuration can be made.

The Universal Station Group and Detail Displays show, which state a point, is in. From the Detail Display the operator can change the state by selecting the parameter *Point Execution Status*, PTEXCST.

2.3.5 Relationship of LMM Point Database to LC Register Module Database

NOTE

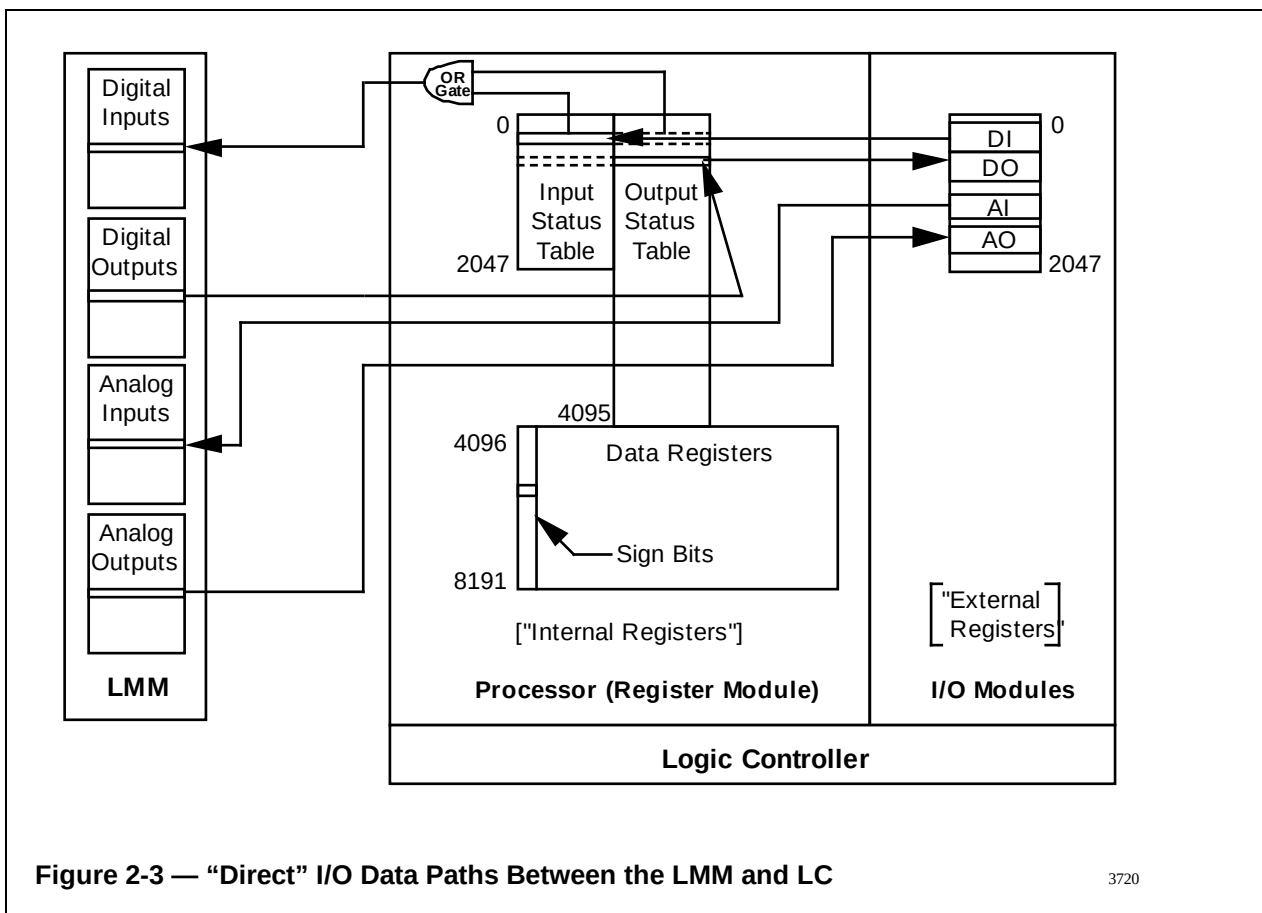
The ladder logic within the LC has primary control over the processing of data written into and read out of its memory. Consequently, **configuration of the LM data points must be coordinated with the ladder logic** if unexpected results at the Universal Station are to be avoided.

All parameters associated with the LM data points are maintained in tables within the Logic Manager Module, based on Tag.Parameter assignments. For purposes of data transfer between the LMM and the Logic Controller, Logic Controller addresses must be specified during data point configuration. The following paragraphs provide an overview of this interface. A detailed description of the interface for each type of data point is discussed later.

As shown in Figure 2-3, the Register Module in the 620 processor contains a total of 8192 (0-8191) data storage locations. Addresses 0 through 4095, known as the *I/O Status Tables*, are for single-bit storage. Addresses 4096 through 8191 are 17-bit *Data Registers*. Digital information is sometimes stored in the sign bit of the Data Registers.

Note that the I/O modules are also assigned addresses 0-2047. These can contain either digital or analog data, depending on the types of I/O cards that are inserted into the I/O slots.

Digital inputs are scanned by the Logic Controller and placed in the corresponding addresses of the Input Status Table, where they are available for transfer to the LMM. Digital outputs are placed in the Output Status Table, by either the ladder logic or the LMM, and are concurrently sent to the digital output cards. Addresses 2048 through 4095 in the Output Status Table are used by the ladder logic for additional single-bit storage. Such status locations are often referred to as *internal coils*.



Analog data points can be passed to and from the I/O registers, as shown, by specifying Logic Controller addresses 0-2047 during configuration of the Logic Manager (assuming that analog cards occupy the addresses used). Of course, addresses 4096-8191 can also be used where interface with the LC ladder logic is desired.

NOTE

The LMM transfers information to/from the Logic Controller by way of a transfer table in the LMM. This table has a 2000-entry limit. **To avoid potential error conditions**, it is important to understand the limitations and operation of this transfer mechanism, which are explained in the following paragraphs.

The 2000-entry Transfer Table is divided into two sub tables:

- an Input Only Table with 1500 entries limited to reading data from the Logic Controller in the single-bit address range of 0-4095, and
- an Input/Output Table with 500 entries that can be used for reading or writing data to the entire Logic Controller address range of 0-8191.

Figure 2-4 shows the relationship of this Transfer Table to the Logic Controller Register Module database.

An LMM point that is invoked to change a value (write to LC) temporarily requires an entry in the 500-entry subtable to transfer the value—until the Output or Register Table is changed in the Logic Controller (1 LC ladder logic scan time). Once the changed value is transferred, this entry is available for other transfers; therefore, some number of entries must be reserved in this subtable for writing data to the Logic Controller from the LMM. This number is dependent on the maximum number of output changes from the LMM that could occur in a given Logic Controller scan. This number is usually in the 30-50 range, and it is therefore suggested that only 450 entries in the 500-entry subtable be used for reading Logic Controller data.

Since data being read from the LC Data Registers (4096-8191) is transferred by the system mechanism first into the 500-entry subtable, unnecessary storage of LMM-accessed values in the Data Registers should be avoided when possible. Therefore, it is recommended that single-bit information be stored in the I/O Status Tables (0-4095) rather than in the sign bit of the Data Registers. Experience indicates that the 1500-entry subtable rarely, if ever, overflows.

The Transfer Table is NOT used for “reading” LM numeric and flag point data. The flags and numeric use a separate block-read transfer mechanism. Using the LMM to “write” flag and numeric data to the Logic Controller WILL use the 500-entry subtable. All other point types use the 2000-entry Transfer Table when accessing data to and from the Logic Controller. This includes! ADRXXXX references.

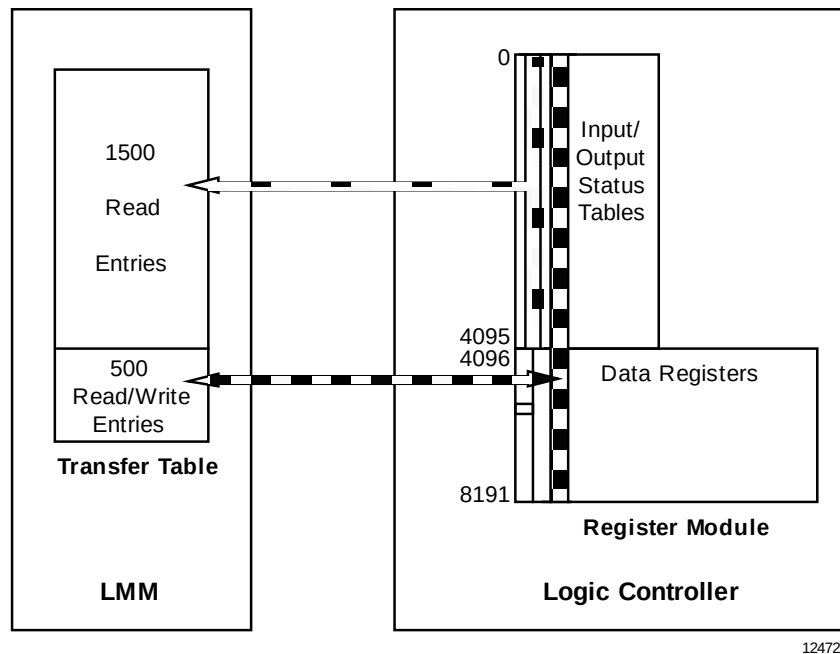


Figure 2-4 — LMM Transfer Table/LC Register Module Table Relationship

Multiple input references to the same Logic Controller address will use only one entry in the table. For example, if the single-bit value “1404” is used for the Contact Cutout (CONTCUT) by several LM points, this entry would only use one location in the table.

Read entries are permanently put in the table until the last reference is no longer required (all point references are deleted).

2.3.6 Point Loading and Checkpointing

The LMM database, like those for most other devices, consists of two basic sets of data:

- Program instructions (fixed for each release of software)
- Variable configuration data
 - “Box”-related (applies to entire device)
 - Point data

Whenever an LM (or other device) is activated—for the first time or after being powered down for any reason—its LMM **program** instructions must be “**LOADed**” from the History Module or

removable media. A simple operating procedure is provided for this purpose. Once loaded, the program remains in tact unless a failure or power-down occurs.

Device or point **data**, on the other hand, is unique for each device or point. Furthermore, some of it can change or be changed by the operator at any time. For this reason, provision is made to automatically **SAVE**, or **CHECKPOINT** the data on a periodic basis. The operator can also request a manual save at any time. If the data is then lost from device memory due to a failure, it can be **RESTORE**d from the checkpointed database.

Provision is also made for saving (*uploading*) the ladder logic program from the Logic Manager onto the History Module or floppy/cartridge disk, and *downloading* it back to the Logic Manager. This is strictly a **program** archiving function.

Refer to the *Logic Manager Implementation Guidelines* in this binder for additional details relative to point loading and checkpointing.

2.3.7 Point Form

Within the system, certain process-related information is grouped together and given a name for purposes of identification. For example, all data pertinent to an input from the process (value, limits, engineering units, etc.) is stored in a database file and assigned a unique point tag name that enables the system to access all or part of that point's data whenever required.

To allow some user flexibility, the Logic Manager, supplies a configurable parameter called PNTFORM (Point Form) that allows the user to define which points are to be used as the primary operator interface for point data. The PNTFORM parameter provides the user with two choices for point form: "Full" and "Component". Points used for the primary operator interface are configured as having "Full" point form, which includes descriptor data and alarm-related parameters.

Points that are configured as having "Component" point form do not require descriptor data and alarm-related parameters; therefore, this type of information is suppressed for "Component" points. This point form should be used for points that provide inputs to the "Full" point, and for those points that handle the outputs from the "Full" point. "Component" points should be used as part of the "Full" point that has been designated a primary operator interface point.

NOTE

The maximum number of points per Network Interface Module (NIM) is 8000. Both full and component points should be counted when checking against this limit.

CAUTION

After initially configuring the database, do a demand checkpoint (SAVE DATA) to checkpoint the configured data.

If the database is not loaded from the checkpoint, then during a cold startup all the point's parameters are set to their default condition. In the case of the full/component selection, a point configured as "full" could be defaulted to the "component" type if the database has not been reloaded from the checkpoint.

The following paragraphs provide examples of the usage of the "Full" and "Component" point forms.

Figure 2-5 shows a single-output single-input digital composite point (MTR100) that interfaces a motor control circuit through digital output (DO) point MTR100OP, and digital input (DI) point MTR100FB. In this example, MTR100 is the primary operator interface and would be configured as a "Full" point. MTR100OP and MTR100FB are parts of MTR100 and would be configured as "Component" points.

Note that assigning tag names in this example for the standard digital input point and digital output point is optional. These points can be alternatively referenced, using the hardware reference addresses that would be assigned as the source and destination on the configuration form for the digital composite point.

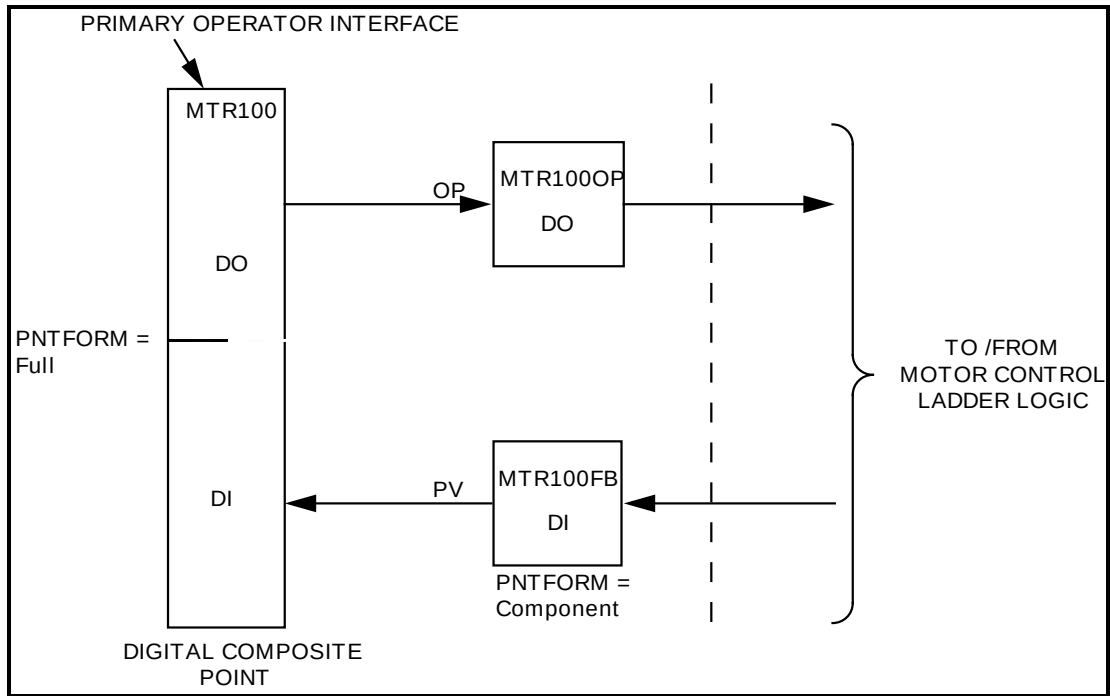


Figure 2-5 — Motor Control Example

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Figure 2-6 shows three types of stand-alone points. FT102 and CT105 are points that could be used for data acquisition; in this case, these points would be configured as full points. Point SW103 is a stand-alone digital output (DO).

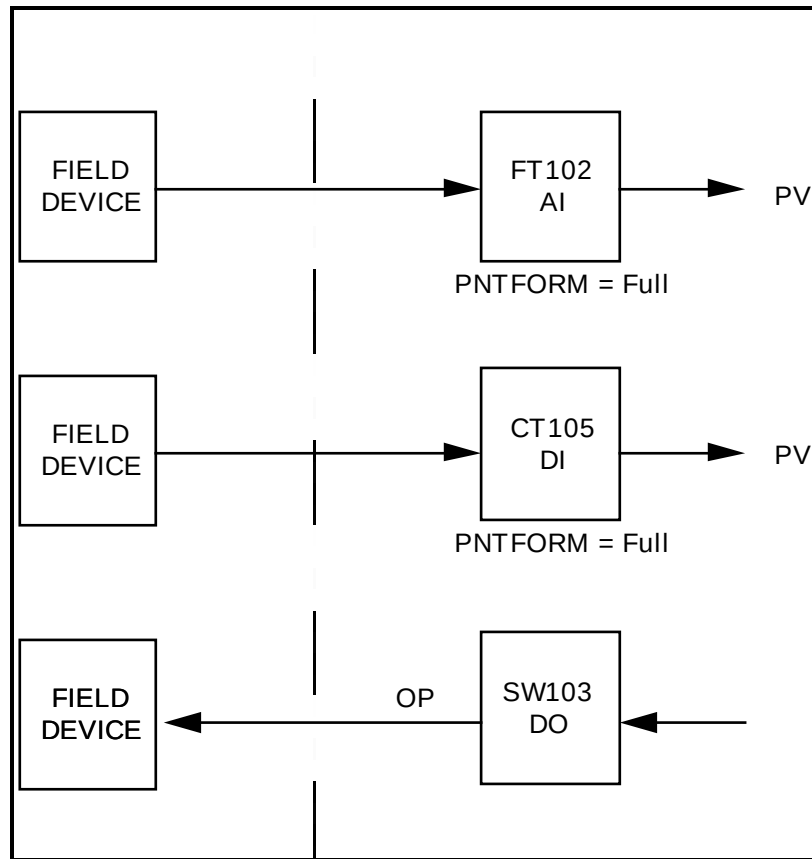


Figure 2-6 — Stand-Alone Points

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2.3.8 Alarming

Alarm detection and reporting for a point can be configured only if the full point form is specified.

2.3.8.1 Alarm Priorities

For each point, a separate alarm priority can be specified for each alarm (for example, PV high alarm can be low priority but PV high high alarm can be emergency). The following alarm priorities are supported for LM points:

- Emergency

- High
- Low
- Journal
- NoAction

Alarm priority configuration information is maintained by the NIM.

2.3.8.2 Alarm Enable Status

Alarm enable status is applicable to full point forms and allows the user to enable, disable, and inhibit alarms. This function is accomplished through the ALENBST parameter. The alarm enable status function is resident in the NIM.

2.3.8.3 Contact Cutout

The main purpose of the contact cutout function is to prevent a proliferation of alarms from being reported to the operator. This function can be used to cut out alarms on a point when they are generated because of alarm conditions that have been detected at other points. Contact cutout is provided for all the point types in the LM and is implemented through the CONTCUT parameter. When the contact cutout state is ON, the alarms at the point are cut out; any new alarm detected is not reported on the alarm summary display of the Universal Station. The alarms continue to be reported to the journals and to the AM or CM60 for event-initiated processing (EIP).

2.3.8.4 Process Alarm Rebroadcast

With R660, acknowledged and unacknowledged process alarms (event-type 21) are rebroadcast if they have not returned to normal (RTN) within the re-alarmed interval specified by the \$REALARM parameter. The re-alarm feature is applicable for all types of process alarms that can have an RTN event.

Re-alarmed is consistent across all stations in the console in that area. Rebroadcast alarms are also sent to the History Module with a new timestamp, and reported to the RTJ and Process Alarm Journal where they will appear as an alarm.

On the alarm displays, a rebroadcast alarm behaves like a new alarm but has a yellow time stamp.

Rebroadcast alarms appear on the GUS Alarm Summary, and on the following GUS/US native-window displays:

- Area Alarm Summary
- Unit Alarm Summary
- Alarm Annunciator
- Point Detail

The re-alarm feature is implemented using the \$REALARM parameter that has a range of 0 to 9999 minutes. The value specified for this parameter determines the amount of minutes that will elapse before the specified point is re-alarmed. Zero is the default value for the \$REALARM parameter. A zero value also inactivates the re-alarm feature.

The user-configured re-alarm interval is the same value for all process alarms associated with the point.

Point types that use the \$REALARM parameter

The following LM point types contain the \$REALARM parameter:

- Analog Input
- Digital Input
- Digital Composite
- Flag

2.3.8.5 Modify \$REALARM

The following scenarios describe the change in the alarm handling because of the change in the re-arming interval while the point is in alarm.

NOTE

To prevent unusual or inconsistent re-arming behavior, do not modify \$REALARM while the point is in alarm.

Scenario 1: Re-alarm interval is modified.

If the re-alarm interval is changed (a non-zero value changed to a different non-zero value) while the point is in alarm, the change is not detected for the current alarm. The change in the interval is detected only for the new alarms subsequently raised for that point. In addition, each alarm for that point is updated with the new interval. Therefore, all alarms of the point are synchronized.

Scenario 2: Re-alarm is cancelled.

If the re-alarm interval is changed (a non-zero value changed to zero) while the point is in alarm, the change is immediately detected and the rebroadcast of the alarms does not occur.

Scenario 3: Re-alarm is initiated.

If the re-alarm interval is changed (zero value changed to non-zero) while the point is in alarm, the change is not detected for the current alarm. The change in the interval is detected only for the new alarms subsequently raised for that point. However, the alarms existing prior to the modification are not rebroadcast.

2.3.8.6 Multiple alarms on the same point

Multiple alarms on the same point are rebroadcast at the same time. The first alarm occurred for that point determines the schedule for all the subsequent alarms. Therefore, all re-alarms for a point appear together in the alarm displays and journals.

2.3.9 Red Tagging

A point can be "red tagged" to indicate that it requires maintenance. This is accomplished by setting the REDTAG parameter to ON. Typically, the operator sets the output of the point to a desired safe value before putting on the red tag. Once red tagged, the mode, mode attribute, external mode switching state, and output cannot be changed. The point can be reconfigured or deleted if the red tag is ON. Red tagging is supported for the digital output, analog output, and digital composite point types. A point must be configured for the full point form in order to be red tagged.

The mode and the mode attribute of the point must be changed to MAN and OPER, respectively, and the external mode switching state, if configured, must be Disabled before the point can be red tagged. To red tag a digital composite point, the output of the point must not be configured for the momentary state.

WARNING

The Red Tagging feature only affects the functionality of the point within the Logic Manager Module only. The PLC address (digital output, analog output) or addresses (digital composite) that a point is connected to within the PLC will NOT be frozen when the point is red tagged (REDTAG parameter is ON). For digital composite points, REDTAG being ON will NOT prevent the override interlocks from changing the OP parameter (the OP parameter represents the status within the PLC and the Red Tagging feature does not affect the PLC and its override logic). The point within the PLC must be disabled before the point is Red Tagged in the Logic Manager Module.

2.3.10 Mode and Mode Attribute Selection List

This R630 function allows an engineer to remove the MODE/MODATTR selections from the operator's standard-display MODE selection lists. This prevents the data point from being placed in the wrong mode and from having the wrong mode attribute assigned.

This function is implemented by the user configuring a mode selection list using the \$MODESEL parameter to determine the applicable modes and mode attribute. During normal operation, the mode selection list is displayed on the standard displays (Detail, Group, Standard Change Zone, Enhanced Change Zone, GUS Faceplate, GUS Group, and GUS Change Zone) when the mode is selected.

The operation of the MAN/AUTO/NORM keyboard keys is also affected by the mode selection list choices.

However, changes to the control mode parameters using AM/CL and custom schematics remain unaffected.

The \$MODESEL parameter can be modified from a custom schematic with Engineer key level access or from AM/CL.

Any change made to the \$MODESEL parameter affects the value of MODEAPPL, i.e., if \$MODESEL is configured to disallow the AUTO mode, then the corresponding boolean in the MODEAPPL parameter array will be reset to FALSE. The Mode attribute entry (PROG) in the MODE selection list can be disallowed by setting \$MODESEL(5) to FALSE. The allowable modes that can apply to a point are dependent on the point type. For the regulatory control points in an AM, the allowable modes are Man, Auto and Cas. Any of these modes can be configured for exclusion from the mode selection list.

The allowable modes that can apply to a point are dependent on the point type as shown in the table below.

ANALOG OUTPUT	DIGITAL OUTPUT	DIGITAL COMPOSIT E
Man Cas	Man	Man

By default, all the modes and mode attributes applicable for a point are allowed in the mode selection list.

NOTE

If parameter EXTSWOPT (External Switch Option) is enabled, its restrictive qualities will take precedence over \$MODESEL.

3. DATA POINT DETAILED DESCRIPTIONS

This section describes in detail each of the nine operator-displayable data point types, including their major parameters, and how the points relate to the Logic Controller (ladder logic) database.

3.1 DIGITAL INPUT POINT

A digital input point converts raw Logic Controller I/O status into UCN-compatible Boolean formats and performs Off Normal alarming. It represents any single LC I/O status point.

Inputs from digital input modules are scanned by the Logic Controller during each Input Status Scan and placed in the Input Status Table. Single-bit data can also be stored in the Output Status Table and in the sign bit of an otherwise unused Data Register location. Note that when data is pulled from addresses 0 through 2047 of the I/O Status Tables, the result is obtained by performing a logical OR on the contents of the Input and Output Tables for the specified address. See Figure 3-1.

The current state of the PV input is represented on the Universal Station Group and Detail Displays as two boxes, as shown in Figure 3-2. The boxes are lit or extinguished depending on the current state of PVRAW and the input direction as configured through the INPTDIR parameter, as shown in the chart in Figure 3-2.

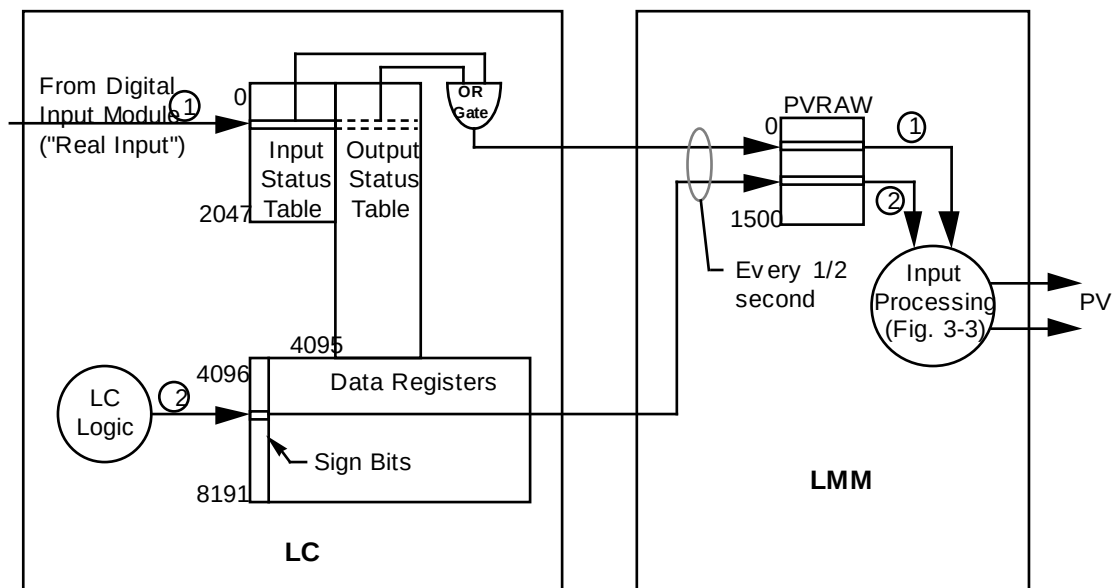


Figure 3-1 — Digital Inputs Data Paths Between the Logic Controller and LMM

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Processing performed on a digital input point is illustrated in Figure 3-3.

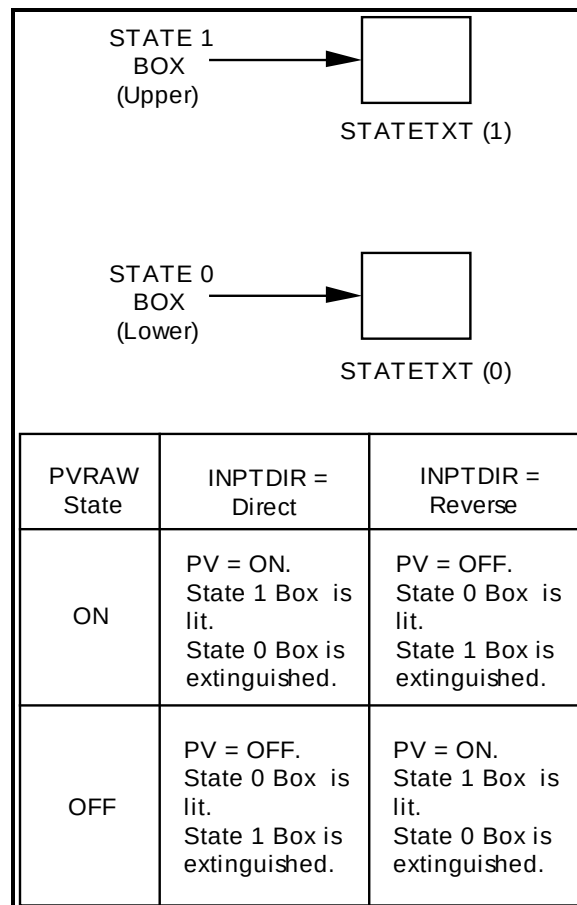


Figure 3-2 — Input States

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The PVAUTO value represents the state of the input signal after the direct/reverse conversion is performed. The digital input point can be configured for PV source selection, detection of off-normal alarms, and for reporting any PV state changes to the system.

3.1.1 PV Source Selection

The PV source parameter (PVSOURCE) option determines the source of the PV for a digital input point. The source can be the PV input from the LC (PVauto), the PV state entered by the operator (PVman), or a user program can supply it (PVsub). If PVSOURCE is PVauto, PV tracks PVRAW.

3.1.2 Off-Normal Alarming

Enabling, Disabling, and Inhibiting Off-Normal Alarms—The ALENBST parameter allows the operator to enable (permit), disable, or inhibit the off-normal alarm. Disabling the alarm still allows the alarm to be listed on the Alarm Summary Display. Inhibiting the alarm prevents the current PV state from being compared to the configured normal state.

Off-Normal Alarming and PV Change Reporting—Off-normal alarming can be selected for the digital input point through the ALMOPT parameter. An off-normal alarm is generated when the input PV state is different than the configured normal (desired) state for the point as specified by the PVNORMAL parameter. The priority of the off-normal alarm is determined through the OFFNRMFL parameter.

Additionally, all PV state changes can be historized as determined by HM volume configuration.

Alarm Delay—When off-normal alarming has been configured and an off-normal alarm is detected, the event is reported to the system. Further off-normal alarms for the same data point are not reported until the time delay (0 to 60 seconds as specified by the DLYTIME parameter) expires. When the time delay expires, the time delay function is disabled and the off-normal alarm for the data point can again be reported.

3.1.3 Event Reporting

The EVTOPT parameter for the digital input allows the user to optionally specify the tag name (EIPPCODE) of a data point in the system that is to be notified when the PV changes state, and/or specify that a time stamp be added to the reported PV state change. For a status input point, EVTOPT has the four possible entries: None, EIP, SOE, and EIPSOE. EIP specifies that the user supply the tag name of the data point in the system that is to receive the PV state change, while SOE specifies that a time stamp be added to the PV state change to establish a sequence of events.

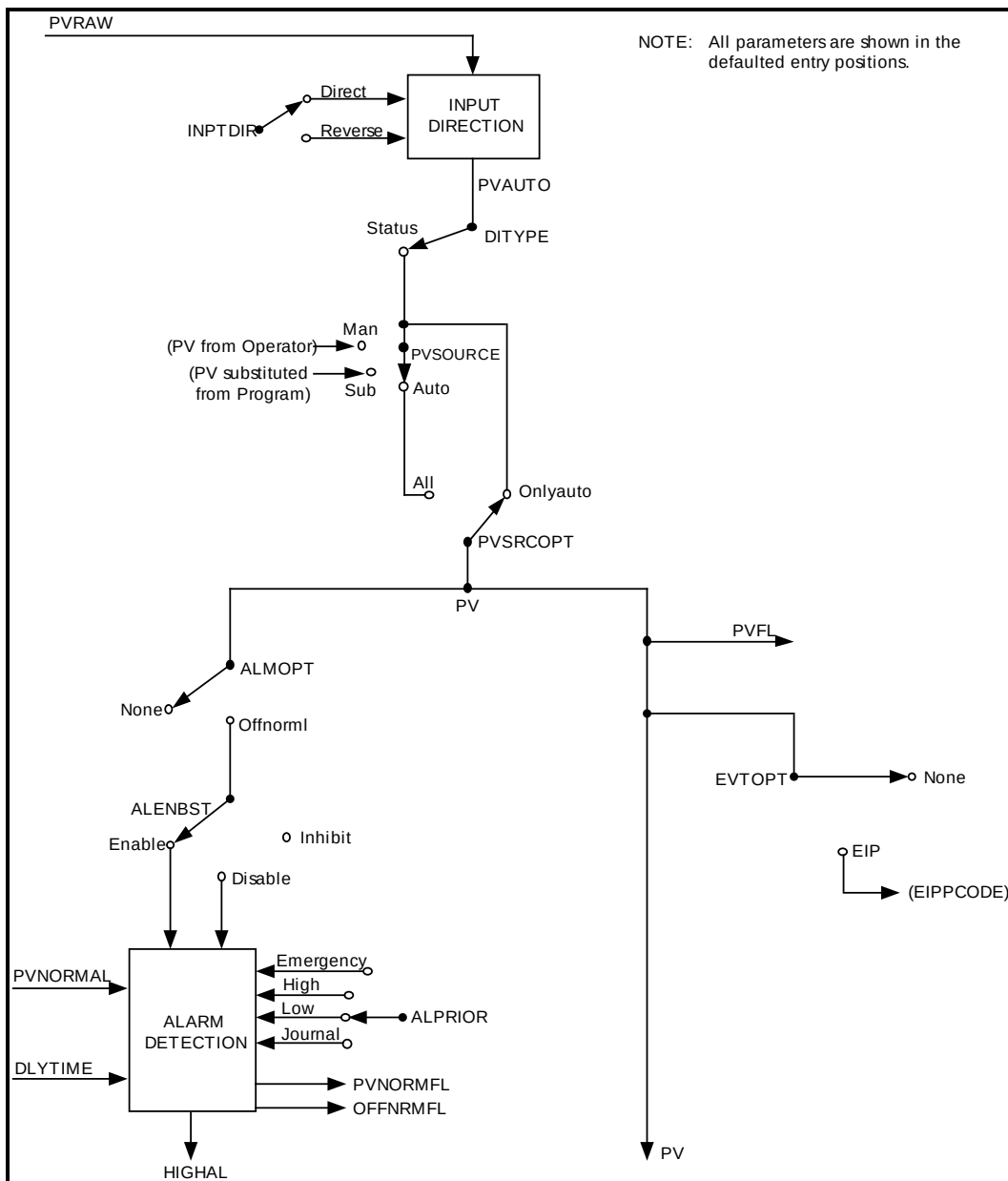
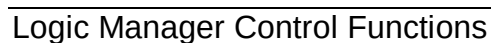


Figure 3-3 — Digital Input Point Processing

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The digital output point is written into a configurable status address in the Logic Controller. This input is processed by the Logic Controller, and the resulting LC output status is converted back to UCN formats where it is stored by the LMM as the “current status” for that point (Figure 3-4). To avoid the appearance of toggling, one LC scan is skipped before again processing the raw input data.

If SO is received from a logic slot, the SO output of the digital output point tracks the SO output provided by the logic slot.



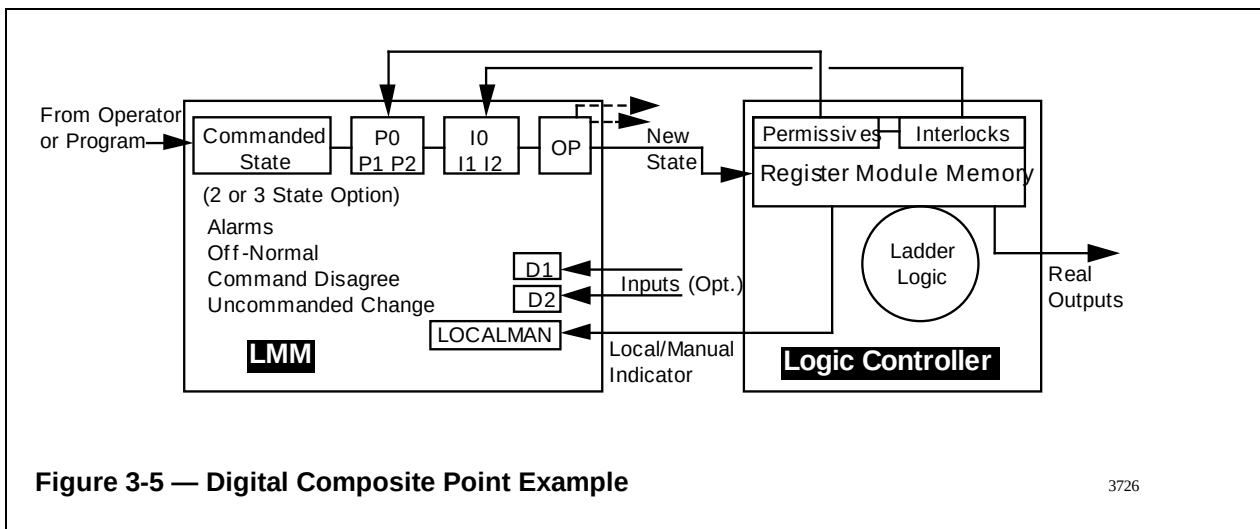
3.3 DIGITAL COMPOSITE POINT

3.3.1 Summary Description

NOTE

For the typical digital composite point to function, **the appropriate algorithm must be programmed into the Logic Controller**. The LMM then accesses the LC points needed to make the digital composite, multiple input/output display available to the operator.

The digital composite point is a multi-input/multi-output point that provides an interface to a load control algorithm in the Logic Controller. The LC in turn controls a target device such as a motor. The Logic Manager interface allows the operator at a Universal Station to monitor and make inputs to the control scheme through the same types of group, detail, and graphic displays supported for Process Manager digital composite points. Figure 3-5 illustrates a typical flow of data between the LMM and Logic Controller for a DC point.



Some of the functions performed by the LM digital composite point are as follows:

- Passes commanded states provided by the operator or program to the point's outputs for use by the Logic Controller.
- Override interlocks. If used, reflect an override operation built into the ladder logic.

- Because the Logic Controller and its ladder logic can alter the state of the output(s) to the field device so that it differs from the commanded state, the output of the LC is read back into the Digital Composite point so that Universal Station operators can be aware when the actual output state is not the same as commanded state.
- Feedback inputs monitor points in the Logic Controller, local digital inputs, or flags. The resulting input is then checked against the commanded state for Off Normal, Command Disagree, or Uncommanded Change, and alarmed as necessary.
- The status of an LC LOCAL/MANUAL input is monitored. No alarming or control is provided.

3.3.2 Point States

3.3.2.1 Normal States

A digital composite point can have two or three normal states that allow the data point to interface with devices that have two or three operational states. The states of a digital composite point are as follows:

- State 1—this is the first active state
- State 0—this is the inactive (middle) state
- State 2—this is the second active state (State 2 is applicable only when the entry for the NOSTATES parameter is 3, signifying that the data point has three states).

Another optional state is also available—The None state (3.3.2.5 in this manual). It is provided so that users can define an LC output state that, when read back from the LC, can differ from the commanded state, and yet not generate an alarm.

These states can best be described by showing how the digital composite data point is depicted in a Universal Station group display, as shown in Figure 3-6. On the display, a separate box represents each state. The State 0 box is the middle box, the State 1 box is the upper box, and State 2 box is the lower box (the State 2 box does not appear if the point has been configured for two states where parameter NOSTATES = 2).

The state descriptor for each box is an 8-character descriptor and appears at the top of the box. In Figure 3-6, the descriptors are FORWARD (for State 1), STOP (for State 0), and REVERSE (for State 2), which are specified by parameters STATETXT(0), STATETXT(1), and STATETXT(2).

The lower portion of each box is split, with the left box being the indicator for the input PV state, and the right box being the commanded-output state indicator. The condition of the PV input that causes the left box to light is determined by parameters D2D1_00 - D2D1_11. OP being commanded to that state determines the condition that causes the right box to light.

As an example, assume that the digital composite point shown in Figure 3-6 is connected through digital output points and digital input points to a motor control circuit and that the current state of the digital composite point is State 0 (STOP).

- In the example of Figure 3-6, the motor is stopped. Both the PV state indicator and the commanded state indicator in the STOP box are on, indicating that the motor is stopped; the OP% descriptor also shows STOP, confirming that the motor is stopped.

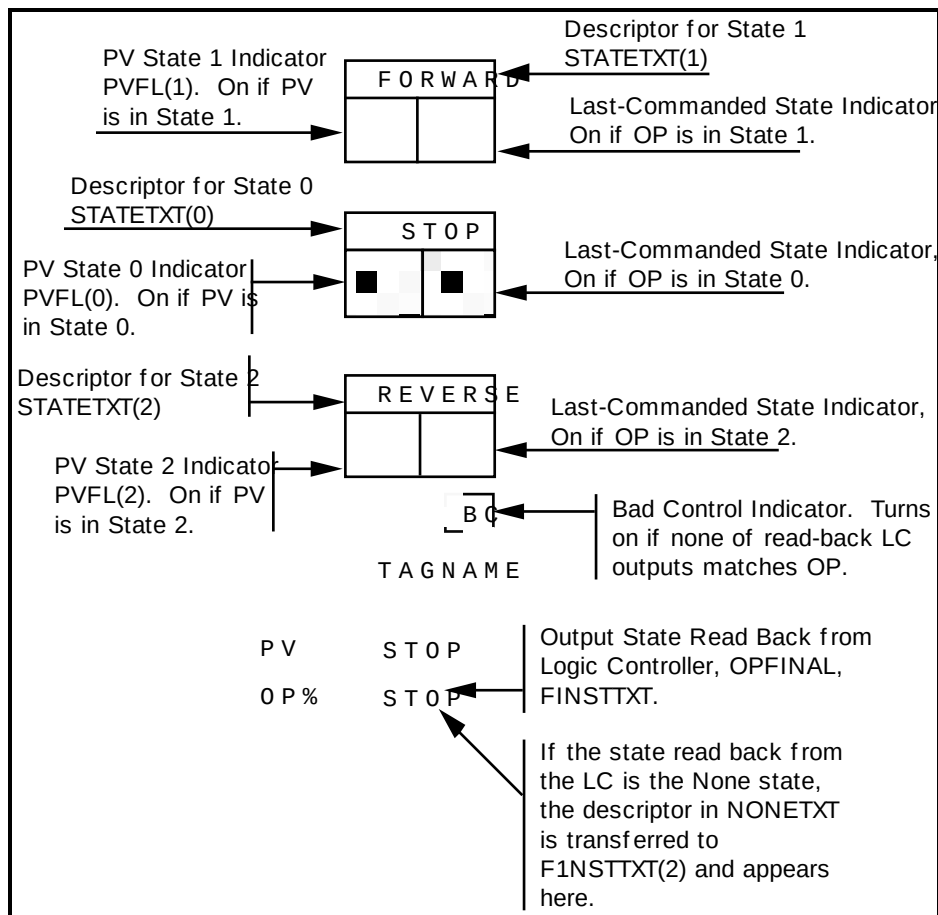


Figure 3-6 — Digital Composite Point Display

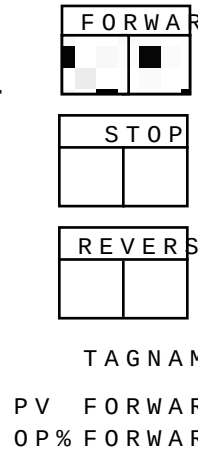
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- When the operator touches the FORWARD box to start the motor running in the forward direction, the commanded output value in OP goes to the state 1 value, lighting the state 1 indicator in the FORWARD box. Outputs from the digital composite point are provided to the LC as determined by the ST1_OPn parameter (where n=1...3).

NOTE

Because the Logic Controller takes a small amount of time to scan through its ladder logic program, the state changes read back from the LC and their indications on the Group or Detail display may take a few moments.

- The state descriptor indicated as the OP% value indicates the state of the output(s) of the motor as read back from the LC. When the read back output in OPFINAL goes to the FORWARD state, this descriptor becomes FORWARD, indicating that the motor is running in the forward direction. If the LC uses the permissives, interlocks, or some other means to alter the output state to something other than the commanded state, that state is indicated by OP%.
- If the PV received from the field goes to the state 1 value, that indicator is lit. If the LC passes the input state unaltered from the field, this also indicates that the motor is running forward. Also, the current state of the PV can be read by a logic slot or by a user program, by accessing parameter PVFL (n) (where n is 0, 1, or 2). If the PV is in State 0, PVFL (0) is on; otherwise, PVFL (0) is off. PVFL (1) and PV (FL2) operate in the same manner for States 1 and 2, respectively.
- Similarly, when the operator presses the STOP box, if the LC passes the output to the field unaltered, the operator is notified that the output to the motor is in the stop state by STOP appearing in the OP% field. The operator can then select the REVERSE box to start the motor running in the reverse direction.



3.3.2.2 Processing of Multiple Outputs

Parameter NODOPTS can be configured to specify one, two, or three LMM outputs from a Digital Composite point, all of which reflect the commanded output state in OP. As described under *Output Connections* (3.3.7), these outputs can be connected to several destinations, including destinations in the Logic Controller. Where there is more than one LC output, the LMM compares each read-back output with OP, and reports the first match in OPFINAL. If no match is found, a bad control alarm is generated (for more information, refer to *Bad Control Alarm* 3.3.13.4).

3.3.2.3 Momentary States

The states of a digital composite point can be configured as being momentary by using the MOMSTATE parameter. The commanded states can be configured as momentary where the state acts like a doorbell (state remains active as long as a switch is pressed). The momentary states (MOMSTATE) parameter specifies which of the states are momentary as follows:

Entry	Description
Mom_1	State 1 is momentary. When released from this state, it jumps back to State 0.
Mom_0	State 0 is momentary. When released from this state, it jumps back to State 1. Mom_0 can be selected only if the NOSTATES parameter = 2.
Mom_2	State 2 is momentary. When released from this state, it jumps back to State 0. Mom_2 can be selected only if NOSTATES parameter = 3.
Mom_1_2	States 1 and 2 are momentary. When released from any one of these states, it jumps back to State 0. Mom_1_2 can be selected only if NOSTATES parameter = 3.

3.3.2.4 Moving/Bad States

The digital composite data point has two states that represent the condition when the current state of the device is "bad" (indeterminate) or the current state is "moving" (from one state to another). The bad state can result when the PV input signals from the process are in an inconsistent state (e.g., for a valve, the limit switches indicating open and closed are both simultaneously on). The moving state is encountered when the device is in transition from one state to another (e.g., a slow moving valve is changing from the open state to the closed state).

The moving/bad state descriptor is located below the State 2 box on the display. In the example described above, the descriptor assigned for the BADPVTXT parameter would be the word FAULTY and the descriptor assigned for the MOVPVTXT parameter would be the word MOVING. These descriptors are configured once for each LM box data point and then are used for all digital composite points in the same LM.

3.3.2.5 The None State

Users can create an optional None state for a Digital Composite point by configuring On in parameter NONECONF. If this state is found in the outputs read back from the LC, the descriptor configured in the NONETXT parameter is placed in OPFINAL and appears on the Group and Detail display as the OP% value (Figure 3-6).

The on and off values that define the None state output values are configured in parameters NONE_OP1 through NONE_OP3, one for each of the three possible Digital Composite point outputs.

If the none state is read back from the LC into OPFINAL, and if it does not match the commanded state in OP, a bad control alarm is not generated. Thus, users can specify an output state (or a pattern of states if there is more than one output), which does not generate a bad control alarm, even though it differs from the commanded state.

3.3.3 Mode Attributes

The digital composite point supports only the manual (MAN) mode with mode attributes (specified by parameter NMODATTR) of **Operator** and **Program**. If the mode attribute is Operator, only the Universal Station operator can provide the commanded output state, as shown in Figure 3-7. On the other hand, if the mode attribute is Program, only user programs (continuous or discontinuous) can provide the commanded output state. The operator mode attribute flag (OPRATRFL) is also provided for potential use by the interlocking logic.

The MODEPERM parameter is provided to optionally prevent the operator (as opposed to the supervisor or engineer) at the Universal Station from changing the mode attribute.

3.3.4 Interlocks

Two kinds of interlocks, permissives and overrides, can be programmed **in the LC ladder logic** for controlling the real Logic Controller output(s) of a digital composite point. When they are so programmed, their positions (ON, OFF) are communicated to the LMM.

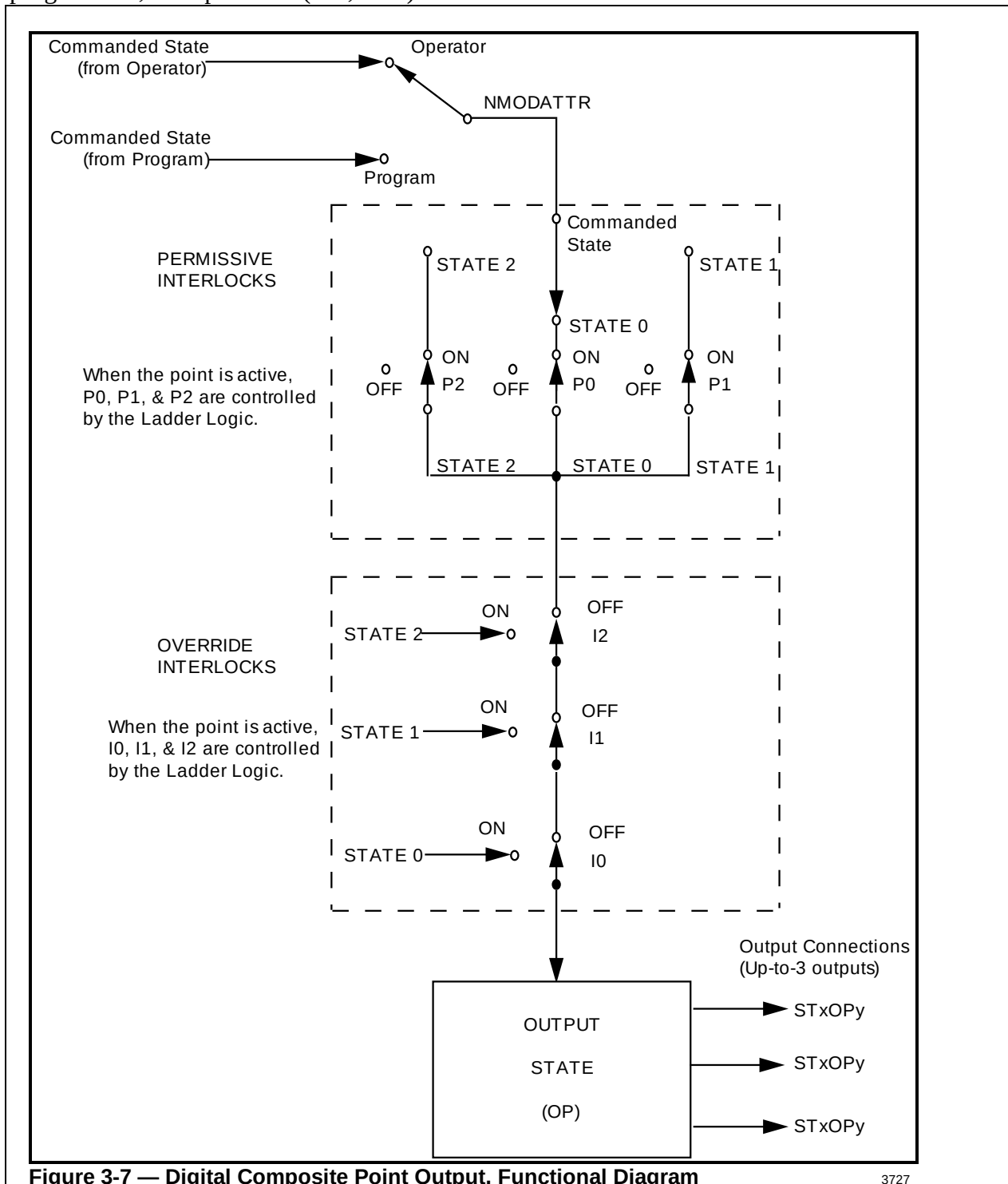


Figure 3-7 — Digital Composite Point Output, Functional Diagram

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3.3.4.1 Permissive Interlocks

The permissive interlock parameters P0-P2, illustrated in Figure 3-7, determine whether the operator and user programs are allowed to change the output of a digital composite point to a specific state. A permissive interlock can be provided for each of the three states. The permissive interlocks themselves never cause the output to change.

For the commanded output to be changed to the desired state, the corresponding permissive interlock parameter must be set to ON. As an example, if a logic slot output has set parameter P1 to OFF, the commanded output of the digital composite point cannot be set to State 1. The permissive interlock parameters P0, P1, and P2 are normally set to ON, thereby allowing permission for all the states. They must be individually set to OFF by the LC ladder logic to prevent access to the corresponding commanded output state.

3.3.4.2 Override Interlocks

CAUTION

We recommend that you program critical interlocks in the Logic Controller ladder logic rather than depending on the LMM's override feature, because this allows the LC to remain in control even if the LMM is placed in the idle state or becomes inoperative.

The LMM override interlock parameters I0-I2 shown in Figure 3-7, which is read in from the Logic Controller memory; force the commanded output (OP) to a specific state regardless of the condition of the permissive interlocks. The operator and user programs cannot change the output state when any override interlock is in the ON condition. An override interlock can be provided for each of the three output states. Note that the actual outputs to the process are supplied only by command from the ladder logic program.

When parameter I0 is set to ON by the ladder logic program, the commanded output state is forced to State 0 (regardless of the condition of parameters I1 and I2). When I0 is OFF and I1 is ON, the commanded output is forced to State 1 (regardless of parameter I2). Finally, when I0 and I1 are OFF and I2 is ON, the commanded output is forced to State 2.

Override interlock parameters I0-I2 are normally set to OFF, disabling all the override interlocks. They must be set to ON (by the ladder logic program) to force the output to go to any specific state.

When all the override interlocks are OFF, the last value of the commanded output is maintained until changed by the operator, program, or another override interlock.

3.3.5 Commanded State (OP)

The command to go to a specific state normally results in the output to the field device going to the commanded state (the Logic Controller can alter output state to the field device). The OP parameter in the digital composite point contains the descriptor for the commanded state. The descriptor is configured in parameters STATETXT (0) through STATETXT (3). The OP parameter is available for configuration only if the number of digital output connections (NODOPTS) is configured to be greater than 0.

Up to nine Boolean parameters of the form STx_OPy (where x = 0, 1, or 3 for the state number and y = 1, 2, or 3 for the output number) allow the user to specify the normal state values that are to be stored by the output connections. For each of the three commanded states per output connection, the user must define the value (ON or OFF) of the state.

For the None state (3.3.2.5), the output values for the three possible outputs is configured in parameters NONE_OP1 through NONE_OP3.

3.3.6 Digital Outputs

Digital outputs for the digital composite point can be configured as either latched or momentary. For momentary outputs, use the MOMSTATE parameter as described in paragraph *Momentary States* (3.3.2.3).

3.3.7 Output Connections

The destinations of the outputs (and the output types) from a digital composite point are specified by the user through the DODSTN (1)-DODSTN (3) parameters. The destinations that can be specified are as follows:

- LC connection
- SO parameter of a digital output point within the same LMM
- PVFL parameter of a local flag point
- LM Box Flag

3.3.7.1 LC Connection

To assign the digital composite point output to an LC connection, the user can enter the following information for the respective DODSTN (n) output connection parameter:

! ADRxxxx

where! ADRxxxx is the address (0-8191) of the point as programmed in the LC.

3.3.7.2 Local Digital Output Point

To assign the digital composite point output to a digital output point, the user can enter the following information for the respective DODSTN (n) output connection parameter:

Tagname.SO

where: Tagname is the 8-character name assigned to the data point through the NAME parameter.

SO signifies the writeable status command of the digital output point.

3.3.7.3 Local Flag Point

To assign the digital composite point output to a local flag point, the user can enter the following information for the respective DODSTN (n) output connection parameter:

Tagname.PVFL

where: Tagname is the 8-character name assigned to the data point through the NAME parameter.
PVFL signifies the PV flag.

3.3.7.4 LM Box Flags

To assign the digital composite point output to an LM box PV flag in the same LM box, the user can enter the following information for the respective DODSTN (n) output connection parameter:

! BOX.FL (nnnn)

or

\$NMhhBxx.FL (nnnn)

where: BOX specifies the same LM device in which the digital composite point resides.
nnnn is the flag number that has a range from 1 to 1024.
hh is the UCN network on which the LM resides.
xx is the address of the LM on the UCN.

3.3.8 Digital Inputs

When configuring digital inputs of the digital composite data point, the user can specify the input connections, PV states, PV source and options, alarming, and change-of-state events. The input portion of a digital composite point can be configured only if the user has entered 1 or 2 for the Number-of-Digital-Inputs parameter (NODINPTS). A functional diagram of the input portion of the digital composite point is shown in Figure 3-8.

3.3.9 Input Connections

The inputs to a digital composite point are specified by the user through digital input-source parameters DISRC (1)-DISRC (2). Inputs to the digital composite point are designated as Input 1 and Input 2, and they can be obtained from any of the following sources:

- LC Connection
- Local Digital Input or Flag Points
- LM Box Flag PV

These sources must be in the same LM box as the digital composite point that is being configured.

The status of Input 1 is represented by parameter D1; Input 2 is represented by parameter D2. Input 2 can be configured only when the entry for the number-of-digital-inputs parameter (NODINPTS) is 2.

3.3.9.1 LC Connection

To assign the PV of a digital input point to the PV input of a digital composite point in the same LM box, the user can enter the following information for the respective DISRC (n) input connection parameter:

! ADRxxxx

where: ADRxxxx is the address (0-8191) of the point as programmed in the LC.

3.3.9.2 Local Digital Input Flags

To assign a local digital input to the PV input of a digital composite point, the user can enter the following information for the respective DISRC (n) input connection parameter:

Tag name.PVFL

where: Tagname is the eight-character name assigned to the logic slot that is providing the output.
PVFL signifies the PV flag.

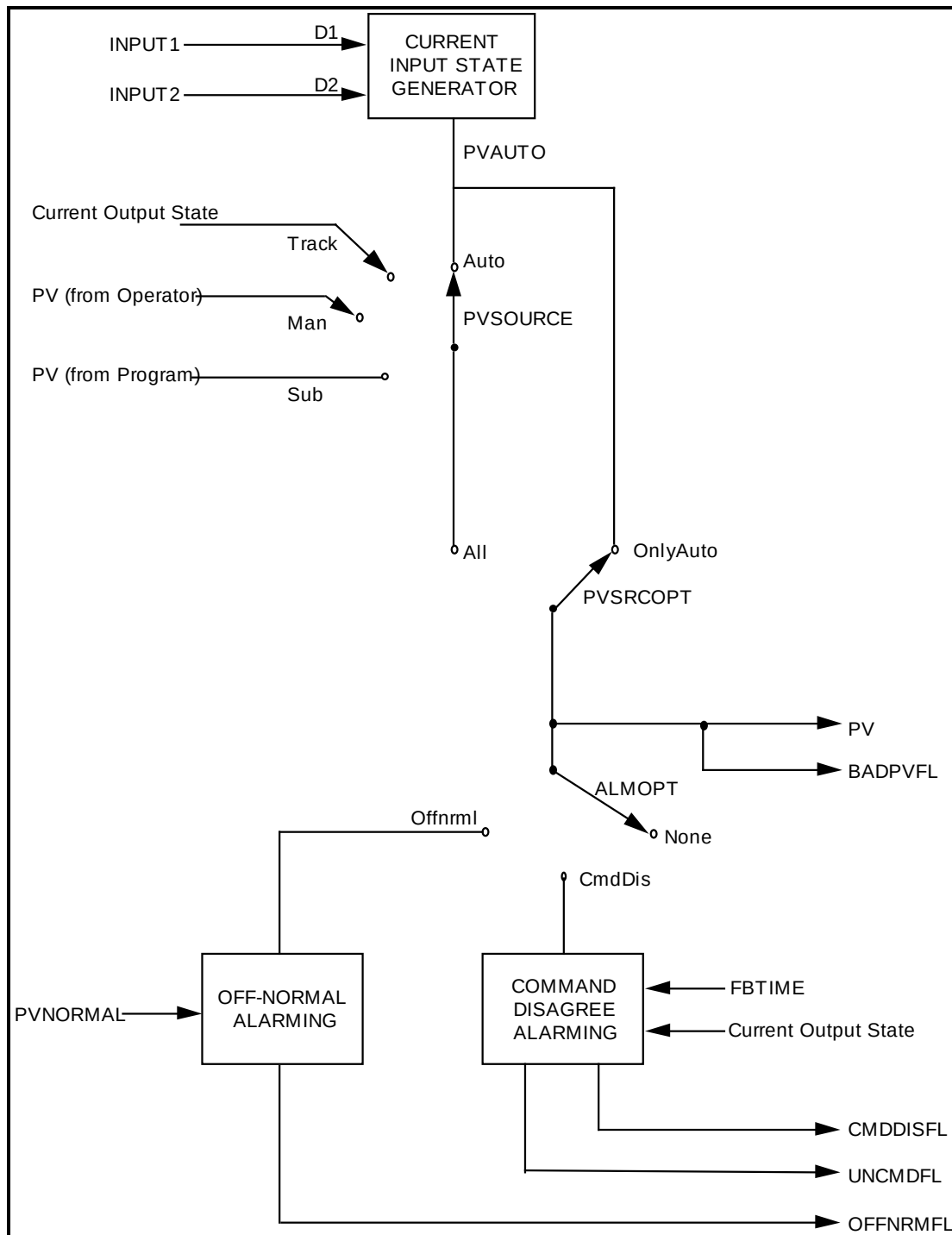


Figure 3-8 — Digital Composite Point Input, Functional Diagram

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3.3.9.3 LM Box PV Flags

To assign an LM box PV flag to the input of a digital composite point, the user can enter the following information for the respective DISRC (n) input connection parameter:

Tagname.PVFL
or
! BOX.FL (nnnn)
or
\$NMhhBxx.FL (nnnn)

where: Tagname is the 8-character name assigned to the box PV flag.

PVFL signifies the PV flag.

! BOX specifies the same LM device in which the digital composite point resides.

nnnn is the flag number that has a range from 1 to 1024.

hh is the UCN network on which the LM resides.

xx is the address of the LM on the UCN.

3.3.10 Current Input State (PV)

The flexibility of the digital composite point allows the user to assign the states of the PV for each possible combination of digital inputs, so that the states correspond to the different applications in which this point type can be used. The PV parameter represents the current state of the interfaced device and is derived from inputs D1 and D2 that can be feedback signals from the process. Separate parameters are used to configure a single-input point and a dual-input point.

NOTE

The inputs to a digital composite point are usually the PVs from digital input points. The digital input points should be configured as component points that force the input direction to be direct (as opposed to reverse). The actual direct/reverse action can be configured by assigning the appropriate PV state to the input as described in the following paragraphs.

For a single-input point, there is only one input parameter (D1). D1 has two possible PV states (PVstate0 and PVstate1) that can be assigned to either of the following input conditions:

D1 = 1 (D1 is ON; parameter D1_1)
D1 = 0 (D1 is OFF; parameter D1_0)

The user has to assign only PVstate0 or PVstate1 to parameter D_1; the system automatically assigns the other PV state to parameter D1_0.

For a dual-input point, there are two input parameters (D2 and D1), which have four possible combinations of input values as follows:

D2D1 = 00 (D2 is OFF, D1 is OFF; parameter D2D1_00)
D2D1 = 01 (D2 is OFF, D1 is ON; parameter D2D1_01)
D2D1 = 10 (D2 is ON, D1 is OFF; parameter D2D1_10)
D2D1 = 11 (D1 is ON, D2 is ON; parameter D2D1_11)

To these four combinations of input values, the user can assign any four of the following five PV states:

- Pvstate1
- Pvstate0
- Pvstate2
- MovPV
- BadPV

PVstates1, 0, and 2 causes the PV indicator to be lit in the respective state box on the group display when the assigned D2D1 input conditions are satisfied. The MovPV and BadPV states cause the respective MOVPVTEXT or BADPVTEXT descriptor to appear below the state boxes on the group display.

3.3.11 PV Source

The PV source parameter (PVSOURCE) determines the source of the current PV state for the digital input portion of the digital composite data point. The possible sources of the current PV state are as follows:

Source	Description
Man (Manual)	Current PV state is provided by the operator from the Universal Station.
Auto (Automatic)	Current PV state is derived from Input 1 (D1) and Input 2 (D2).
Track	Current PV state is the commanded output state.
Sub (Substituted)	Current PV state is provided by a user program.

3.3.11.1 PV Source Option

During configuration, the user can specify the PV sources that can be used for this data point. Parameter PVSRCOPT allows the user to select the PV source as being only AUTO, or to select all the PV sources in the above listing as allowable sources of the PV.

3.3.12 Alarming

The digital composite point can be configured to detect and report command disagree, uncommanded change, or off-normal alarms through the use of the ALMOPT parameter. The user also has the option of specifying no alarming for the data point. These alarm options can be configured only if the digital composite point is configured to have inputs, or inputs and outputs.

3.3.12.1 Command Disagree and Uncommanded Change

This type of alarming is selected by entering CmdDis for the ALMOPT parameter. When the commanded-output state is changed and the actual input PVstate does not change accordingly within a predefined feedback time, a command disagree alarm is generated. The feedback time (1 to 1000 seconds) is specified by the FBTIME parameter.

This alarm condition returns to normal when the input PV state and the commanded-output state are the same. If the commanded state is momentary, no alarm is generated. For example, a motor may have two PV states (RUN and STOP), but there may be three commanded output states (RUN, STOP, and JOG) where JOG is defined as a momentary state. Command disagree alarming is performed for only the RUN and STOP commanded output states.

If a change does not occur in the commanded output state but the input PV state changes (and the PV is not bad), an uncommanded-change alarm is generated. This alarm condition returns to normal when the input PV state and the commanded state are the same. If the point state has been configured as being momentary, this type of alarm is not applicable.

3.3.12.2 Off-Normal

Entering Offnorml for the ALMOPT parameter configures the detection of off-normal alarms. The user through the PVNORMAL parameter defines the normal state of the PV input. When the PV input state is different than the state specified by the PVNORMAL parameter, the off-normal alarm is generated. The alarm condition returns to normal when the PV input state and the specified PV normal state are the same.

3.3.12.3 Change-Of-State Events

Any transitions in the PV input state can be used for triggering event-initiated processing of points in the Application Module (AM). It is configured by entering EIP for the Event-Report-Option parameter EVTOPT. The user must enter the tag name of the AM or CM point using the EIPPCODE parameter.

3.3.12.4 Bad Control Alarm

A bad control alarm indicates that the Logic Controller output state does not match the commanded state in the OP parameter. After a change in the output is commanded by an operator at a Universal Station or by a user-written program, the LMM defers the calculation of the read back status from the LC for one LC scan to allow the LC to update its output to the field. The LMM compares each of the one, two, or three outputs with OP, and reports the first match in OPFINAL. A Bad Control alarm (BADCTLFL = On) is generated if no match is found on any output. This causes the bad control indicator (Figure 3-6) to appear on the Group or Detail display for the point.

3.3.13 Local Manual Indication

Field devices that are interfaced by a digital composite point often have a local HAND/OFF/AUTO (often called HAND/OFF/REMOTE) switch. Unless this switch is in AUTO position, the control system (LM) may not have any control over that device. The users can optionally feedback the switch position into the LM to obtain some display indication for the Universal Station operator.

This indication is provided by the word LOCALMAN appearing at the bottom of the digital composite point on a group display. Further, when in local manual, any changes to the output by the operator, user programs, or override interlocks are typically prohibited (assuming the LC is programmed accordingly). The override interlocks should be programmed to take effect as soon as the local manual condition is cleared.

To support HAND/OFF/AUTO switches, a Boolean flag called LOCALMAN is provided for connection to the LC. The ON state indicates that the switch is not in AUTO position.

Once again, be aware of the importance of assuring compatibility of the ladder logic programming with the connections established for the Logic Manager digital composite point.

3.4 ANALOG INPUT POINT

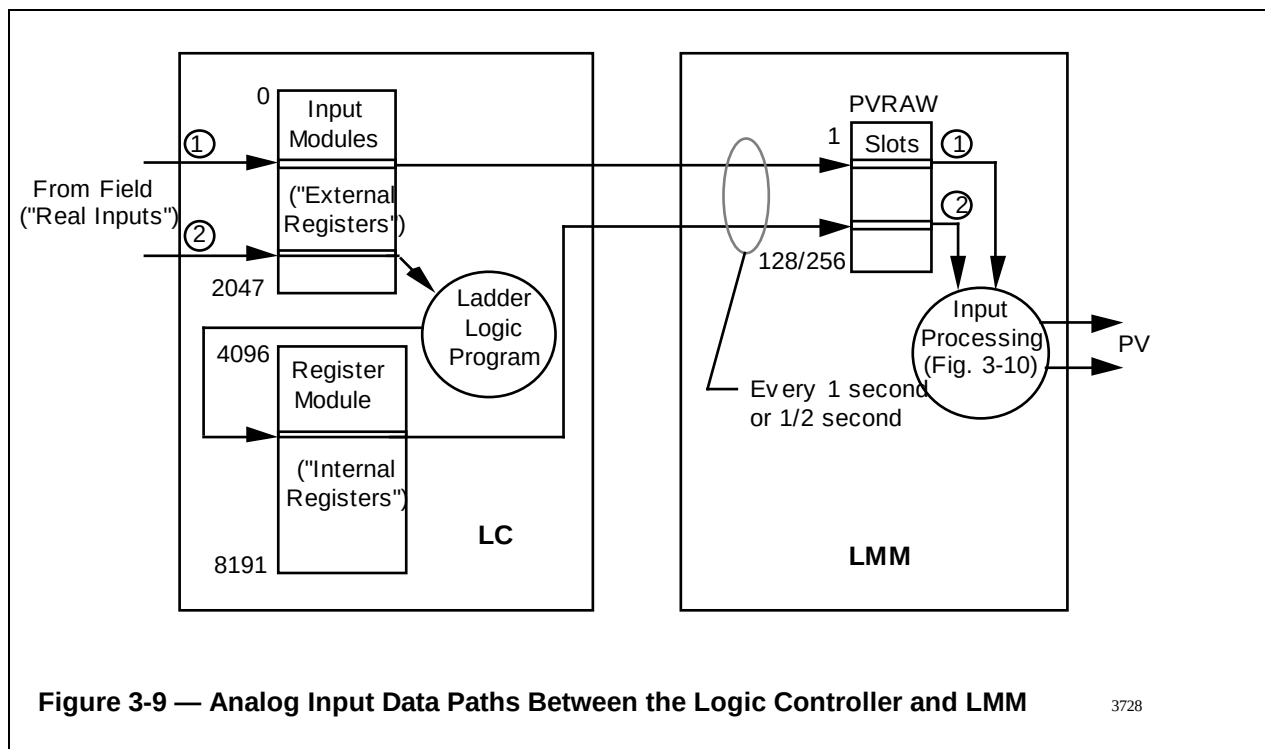
The analog input point converts raw LC numbers into engineering units for use by the rest of the System. Parameter PVFLTPT is configured to designate whether the Logic Controller data format is Signed, Unsigned, or Floating Point. The Logic Manager can be configured to update analog inputs on either a 1-second or 1/2-second cycle.

Figure 3-9 illustrates how analog input points in the LMM can originate directly from a 620 Analog Input Module (1) or can be provided by the ladder logic program (2).

Upon receiving the raw AI value, the LMM performs the following functions:

- PV characterization
- Range checking and PV filtering
- PV source selection
- Alarm detection

These functions and their associated parameters are illustrated in Figure 3-10 and described on the following pages.



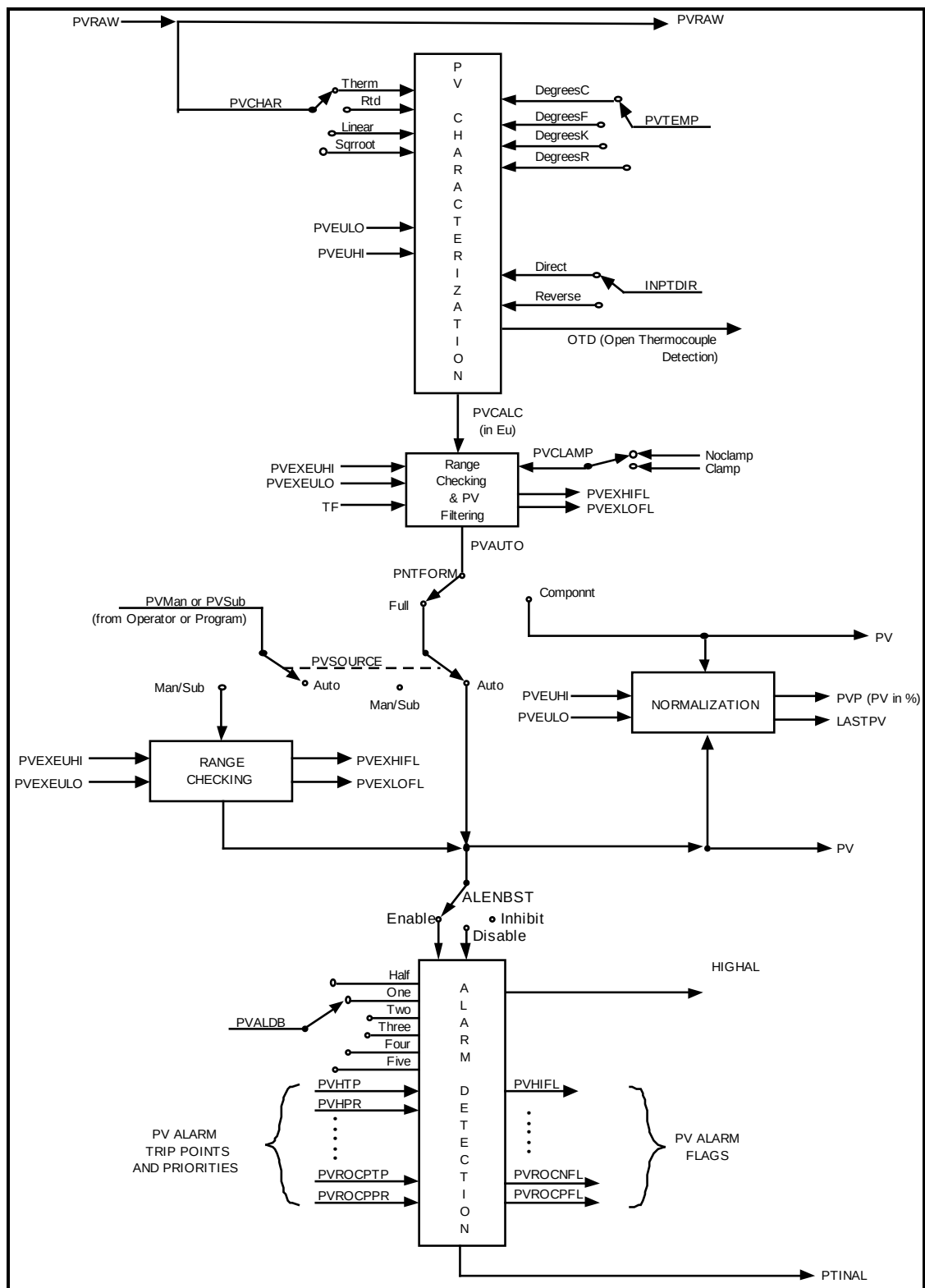


Figure 3-10 — Analog Input Point Processing

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3.4.1 PV Characterization

The PVRAW signal received from the LC is a percentage of the source range. It is characterized based on the entries that the user makes for the PVCHAR, PVTEMP, and INPTDIR parameters as shown in Figure 3-2.

The value is then converted to engineering units. The engineering unit conversions that are performed are listed below, and described in the following paragraphs.

- Linear
- Square Root
- Thermal
- Slidewire

3.4.1.1 Linear Conversion

The PV value is converted to a floating-point number. The output value of the linear conversion is PVCALC. The state of the input direction parameter (INPTDIR) is taken into consideration during the calculation of PVCALC as follows:

When INPTDIR is Direct:

$$\text{PVCALC} = \frac{\text{PVRAW}}{100} * (\text{PVEUHI} - \text{PVEULO}) + \text{PVEULO}$$

When INPTDIR is Reverse:

$$\text{PVCALC} = \text{PVEUHI} - \frac{\text{PVRAW}}{100} * (\text{PVEUHI} - \text{PVEULO})$$

3.4.1.2 Square Root Conversion

The square root calculation is applied to the PVRAW input such that 100% of span = 1.0. The square rooted value is then converted to engineering units based on the configured PV engineering-unit range values (for example, square root of 100% = 100%; square root of 50% = 70.71%). The output value of the square root conversion is PVCALC, which is calculated based on the state of the input direction parameter (INPTDIR) as follows:

If INPTDIR is Direct:

$$\text{PVCALC} = \frac{\sqrt{\text{PVRAW}}}{100} * (\text{PVEUHI} - \text{PVEULO}) + \text{PVEULO}$$

If INPTDIR is Reverse:

$$PVCALC = PVEUHI - \frac{\sqrt{-PVRAW}}{100} * (PVEUHI - PVEULO)$$

3.4.1.3 Thermal Conversion

Thermal linearization is performed on thermocouple and RTD inputs, and is selectable by parameter PVCHAR. The analog input point supports the following thermocouples:

- Btherm
- Etherm
- Jtherm
- Ktherm
- Rtherm
- Rptherm
- Stherm
- Ttherm

Table 3-1 — Thermocouples (Degrees C)

	NORMAL RANGE		EXTENDED RANGE	
TYPE	PVEXEULO	PVEXEUHI	PVEXEULO	PVEXEUHI
Btherm	600	1650	100	1820
Etherm	-150	500	-200	1000
Jtherm	-100	750	-200	1200
Ktherm	0	1100	-100	1370
Rtherm	550	1500	0	1700
Rptherm	550	1500	0	1770
Stherm	550	1500	0	1700
Ttherm	-200	300	-230	400

Table 3-2 — RTDs (Degrees C)

TYPE	PVEXEULO	PVEXEUHI
PtDinRTD	-180	800
PtJisRTD	-180	650
NickLRTD	-45	315
CopprRTD`	-20	250

The analog input point supports the following 3-wire RTDs:

- PtDinRTD
- PtJisRTD

- NicklRTD
- CopprRTD

For an RTD, the lead wire compensation is calculated and then subtracted from PVRAW.

Thermocouple and RTD extended range parameters (PVEXEULO, PVEXEUHI) are defaulted to fixed values according to the sensor types. The accompanying tables show these values in degree C. For other engineering units they are appropriately converted.

3.4.2 PV Range Checking and Filtering

PV range checking ensures that the PVCALC output of PV characterization is within the limits defined by parameters PVEXEULO and PVEXEUHI. If either of the limits is violated, the output of the range check is set to BadPV if clamping has not been specified. If clamping has been specified, the output of the range check is clamped.

If the range-checked and filtered value is less than the value specified by the user-configured LOCUTOFF parameter, the final output called PVAUTO is forced to PVEULO.

First-order filtering is performed on PVCALC, as specified by the user through parameter TF (filter lag time).

3.4.3 PV Source Selection

The PVSOURCE parameter allows the user to select the source of the PV for this data point. As shown in Figure 3-2 the PV can be provided by the Range Checking and Filtering circuit (when PVSOURCE is Auto), or it can be a manually entered PV (when PVSOURCE is Man or Sub).

In addition, the PV source option parameter (PVSRCOPT) determines whether it is permissible to change the PV source to a source other than Auto. PVSRCOPT has two states: OnlyAuto and All. The All state allows the PV to be manually entered for this data point.

3.4.4 Alarming

The analog input data point compares the PV to threshold values and records the alarms in the database of the data point. The Logic Manager Module (LMM) then reports the alarms. The parameters that are associated with alarming in the analog input point are as follows:

ALENBST	HIGHALPR	PVHIPR
BADPVFL	PTINAL	PVHITP
BADPVPR	PVALDB	PVLOFL
CONTCUT	PVEXHIFL	PVLOPR
EIPPCODE	PVEXLOFL	PVLOTP
HIGHAL	PVHIFL	

Refer to the *Logic Manager Parameter Reference Dictionary* in this binder for definitions of these parameters.

3.5 ANALOG OUTPUT POINT

Figure 3-11 shows an overview of the data paths in the Logic Manager for analog output points. The analog output OPFINAL parameter value, as shown in Figure 3-11, can be (1) written directly into a register in an output module or (2) written to an internal register for handling by the ladder logic (2). Analog output points can be configured to be processed on either a 1-second or 1/2-second cycle, although actual output may take place much faster than this, depending on loading of the Logic Controller.

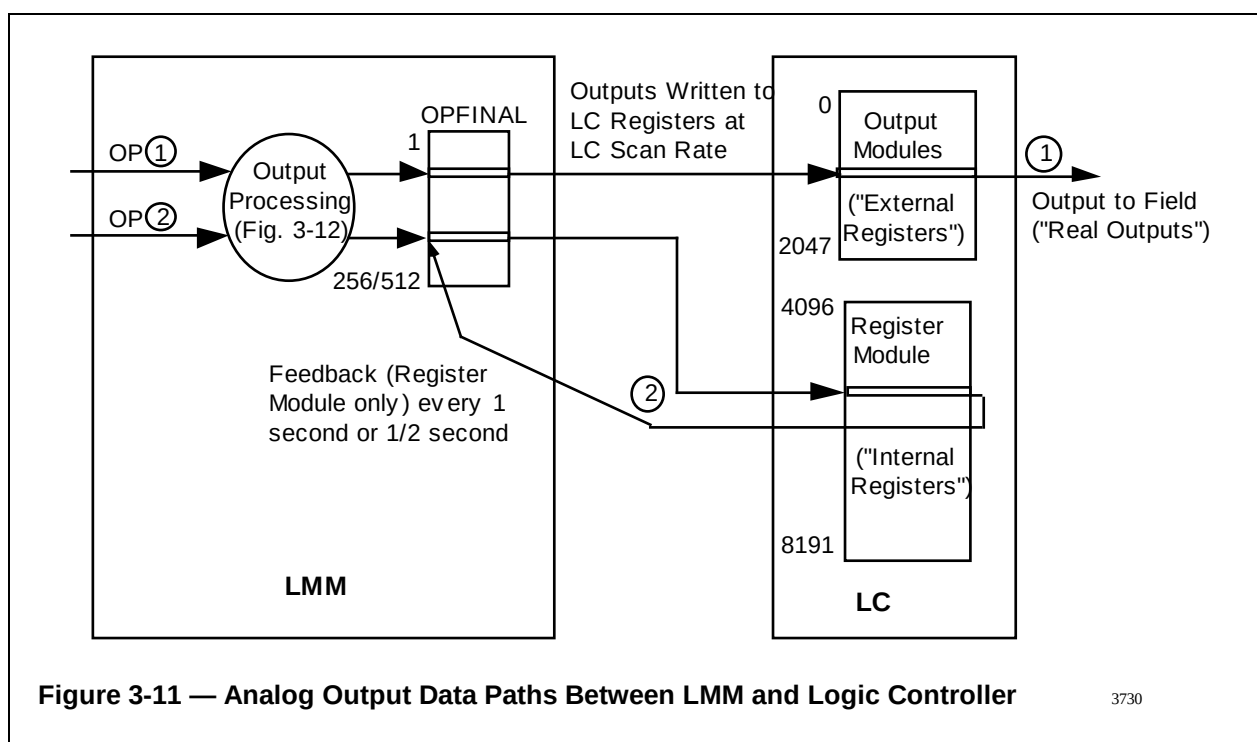


Figure 3-12 shows the point processing parameters that can be configured for the LMM. In addition to the switch options illustrated, the following processing takes place:

- Direct/Reverse Output Function
- LC Data Conversion Function

3.5.1 Direct/Reverse Output

Parameter OPTDIR allows the user to specify whether the output of the data point is direct acting (where minimum value = 0%, and maximum value = 100%) or reverse acting (where minimum value = 100%, and maximum value = 0%).

3.5.2 LC Data Conversion

LC data conversion allows the user to specify the type of output to be supplied to the LC: Unsigned, Signed, or Floating Point. It is implemented by setting parameter PVFLT OPT to the appropriate format.

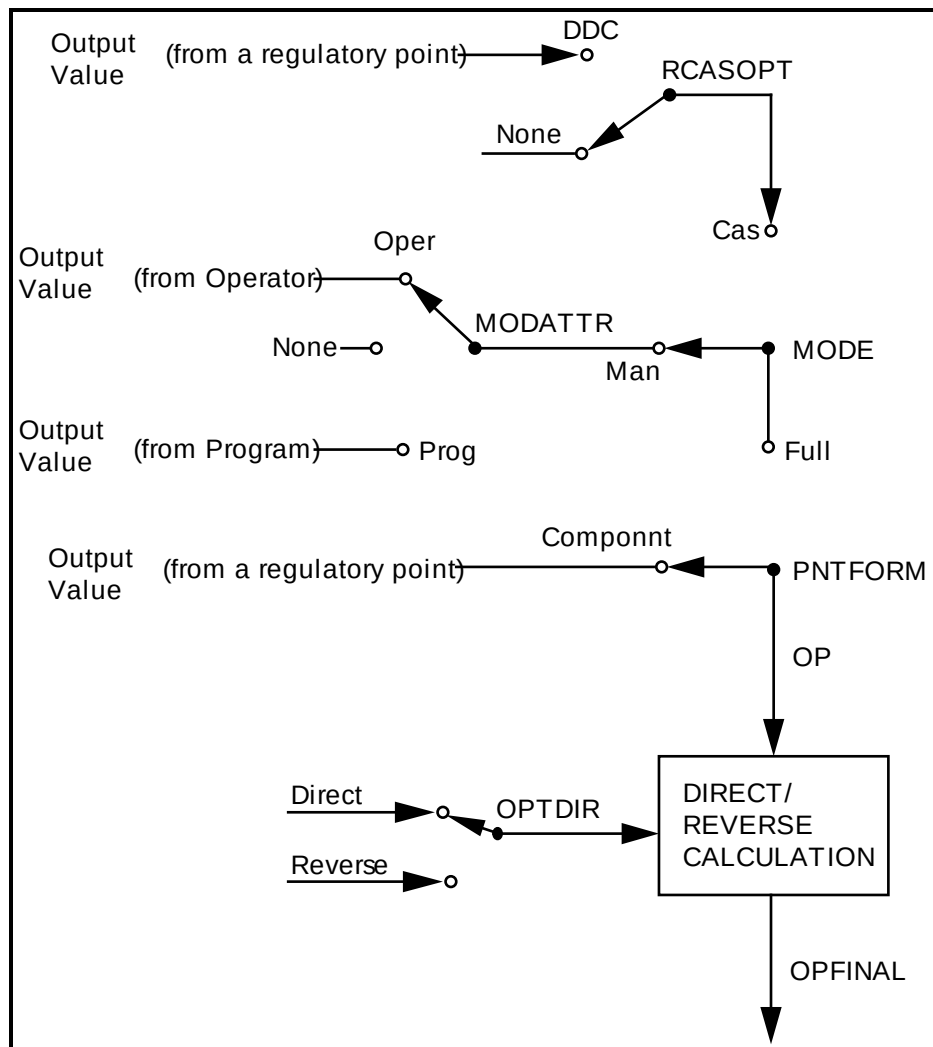


Figure 3-12 — Analog Output Point Processing

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3.6 LINKAGE (LOGIC) POINT

3.6.1 Summary

The function of the LM Linkage point is limited to the transfer of data between UCN and LM connections, any pairing of which is acceptable. Data types can be Boolean, unsigned, integer, or real.

The Linkage point has, therefore, sometimes been referred to as the “linkage point”, and is the basis for the LMM’s application-level support to peer-to-peer communications.

3.6.2 Configuring the Linkage Slot

Each Linkage point accommodates 12 input connections and 12 output connections; with a total maximum of 50 connections scheduled for processing during any single scan cycle.

3.6.2.1 Linkage Input Connections

The user assigns inputs to the Linkage Point during configuration by using logic input connections. The user can specify up-to-12 inputs that are assigned to linkage-slot inputs L(1) to L(12), as shown in Figure 3-13. The linkage input connections are specified through the LISRC parameter.

The inputs to the linkage slot can be obtained from any of the following sources:

- Any Boolean, integer, enumeration, self-defining enumeration, or real parameter within this LMM, or in another UCN node, using the "Tagname.Parameter" format.

NOTE

Integer, enumeration, or self-defining enumeration parameters are automatically converted to real values.

- An address (! ADRxxxx) in the LC.

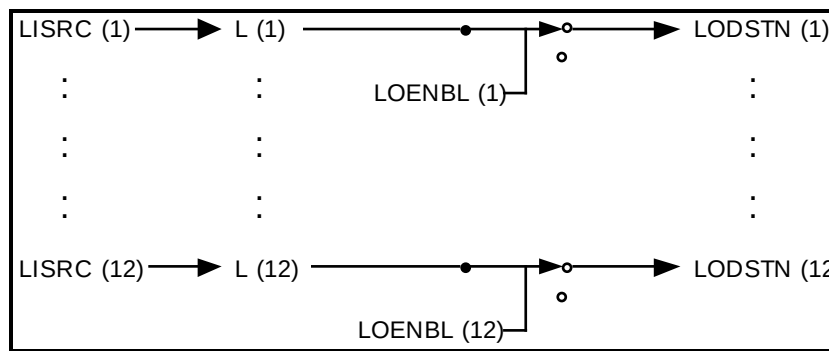


Figure 3-13 — Linkage Slot

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3.6.2.1.1 Effect of Communication and Configuration Errors on Linkage Input Connections

An input obtained from a failed device can result in a communication error. Alternatively, an input can be coming from a digital composite point in another LM and the point mix in that LM can be changed such that the specified point no longer exists. This results in a configuration error. In order for the routine to be able to continue in spite of a configuration error, the following special features are provided:

- **Bad Boolean Inputs**—If a Boolean input is not successfully fetched, its value is defaulted based on the Linkage Input Bad Handling Option parameter (LIBADOPT) as follows:
 - LIBADOPT = ON The ON state is substituted for the unsuccessful input.
 - LIBADOPT = OFF The OFF state is substituted for the unsuccessful input.
 - LIBADOPT = HOLD The previous value (the last successfully fetched value) is substituted for the unsuccessful input. On startup, the previous value is defaulted to the OFF state.
- **Bad Real Inputs**—If a real input cannot be successfully fetched, its value is defaulted to NaN.

3.6.2.2 Output Connections

Linkage output connections are used to write the values of local parameters of a respective linkage slot to the configured destinations. Up-to-12 output connections can be configured for each linkage slot. The destinations are specified by the parameter LODSTN by using the "Tagname.Parameter" format, the UCN hardware reference address format, or the LC address. Each destination address is tied to the input bearing the same number (LODSTN(1) to L1, LODSTN(2) to L2, etc.).

The linkage output connection can write the selected local parameters of a linkage slot to any of the following destinations:

- Any Boolean or real parameter in this LMM, or another UCN node.
- An address in the LC.

Associated with each output connection is a linkage output enable flag, LOENBL(n), which is always an LC address (!ADRxxxx). The LC parameter pointed to by LOENBL(n) must be ON for the corresponding output connection to write to the specified destination. The default value of -1 represents the ON state.

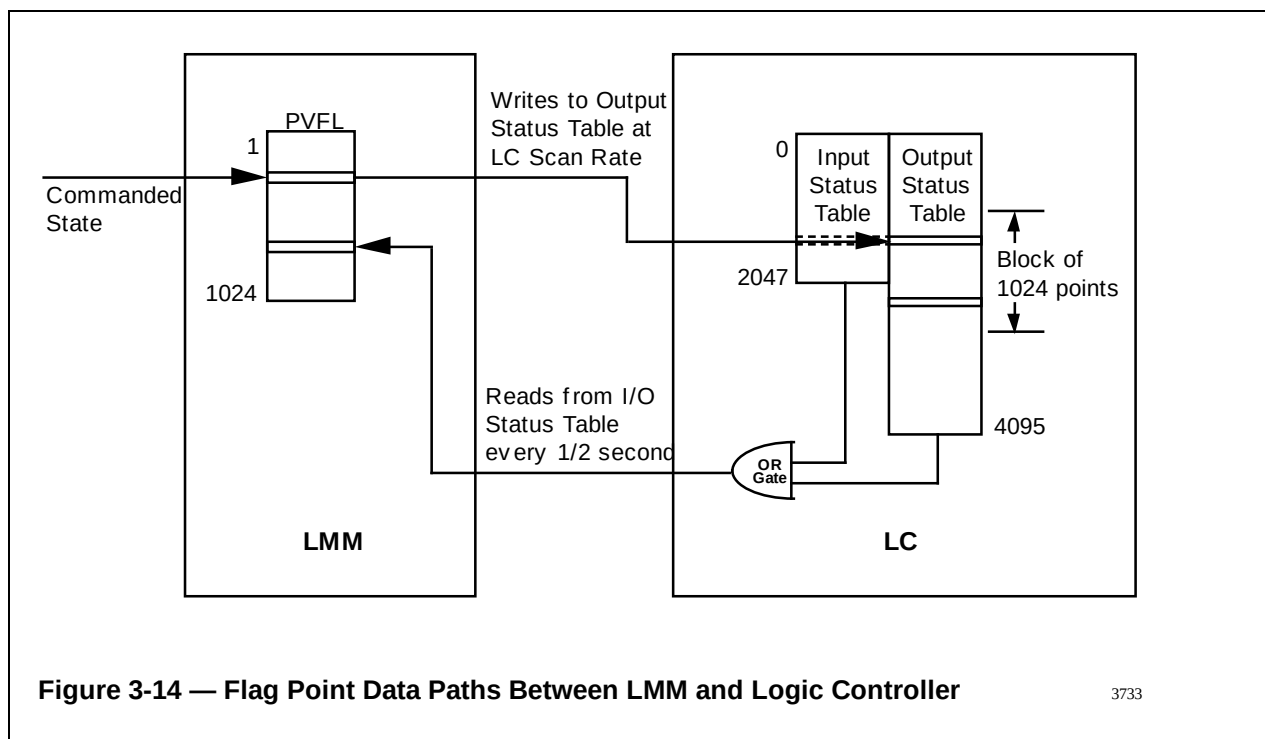
3.6.2.3 Generic Descriptors

Up-to-12 user-defined generic descriptors are provided with each linkage slot for identifying external inputs (L1, L2, etc.) of the linkage slot with custom names that are to be shown on Universal Station displays. The actual number of descriptors used is determined by the NODESC parameter for this linkage slot. The corresponding 8-character descriptor is defined by GENDESC(n).

3.7 FLAG POINT

3.7.1 Summary

A Flag data point is a 2-state (ON and OFF) point that is used for storing a Boolean value. The operator can supply the value by an output connection from another LM point, by any LM or PM on the same UCN, by the LC, or by a node on the LCN. Flag points can be configured as Full or Component points as determined through the PNTFORM parameter.

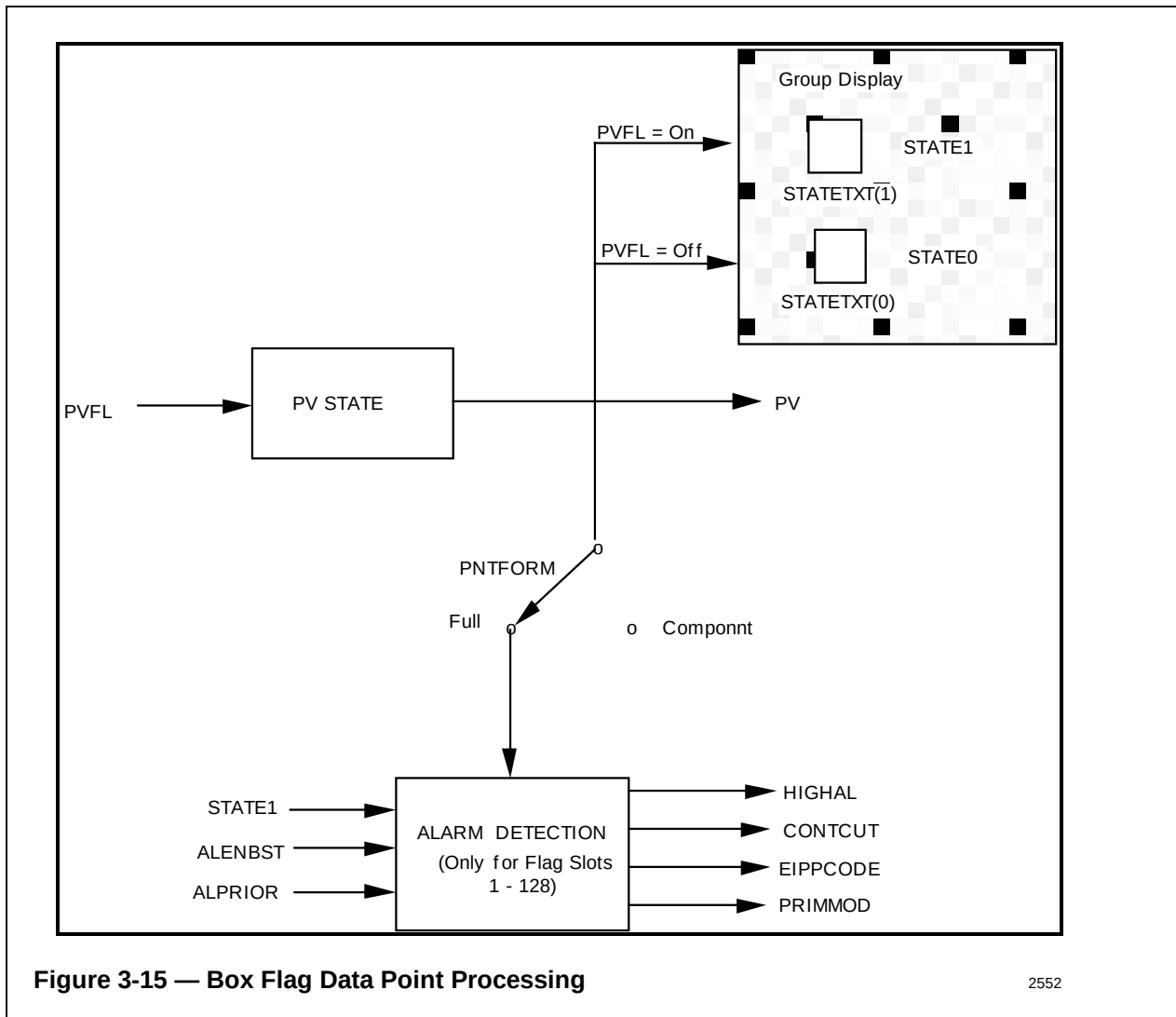


The states of Flag data points can change whenever they are accessed by other system functions such as an operator or a user-written program. Data paths between the LMM and Logic Controller are shown in Figure 3-14.

As shown in Figure 3-15, the input to the flag point is provided by parameter PVFL, which can be ON or OFF. PVFL will set the flag point PV state to the same state as PVFL. The PV is then available as an output from the flag point and other points in the LM and in the system can access its current state.

PVFL will also light the STATE1 and STATE0 boxes on the Universal Station Displays, depending on its state. If PVFL is ON, the STATE1 (upper) box will be lit; if PVFL is OFF, the

STATE0 (lower) box will be lit.



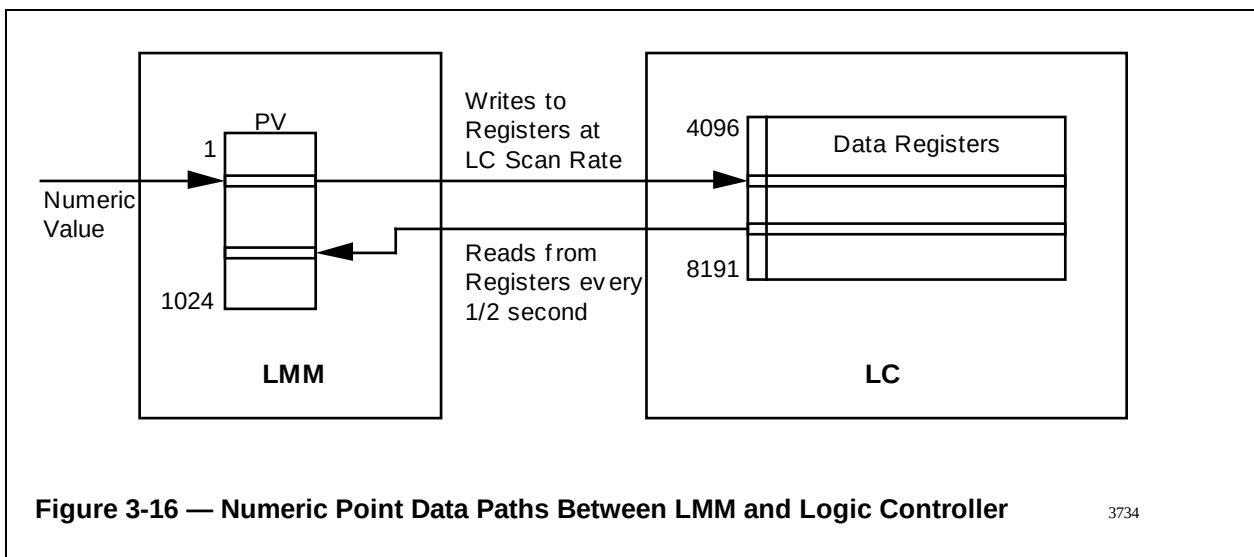
3.7.2 Alarming

The first 128 flag points (slots) can be configured for off-normal alarming. An alarm will be generated when the PV of the flag point is changed from STATE0 (OFF) to STATE1 (ON). Alarming is available only if the flag point has been configured as a full point.

3.8 NUMERIC POINT

Numeric data points are used to store integer values such as operator entries, accumulations, or the intermediate results of program calculations. LM Numeric points can be configured as either full or component points. They are accessible to routines in the same LM, to any LM or PM on the same UCN, to the LC, and to any node on the LCN.

The parameter values of Numeric points can change whenever they are accessed by a system activity, such as by an operator or a system program. Figure 3-16 shows the LM data paths for numeric points.



3.9 TIMER POINT

The Timer data point introduces the capability for automatic timekeeping into Logic Manager activity. This type of data point keeps track of the elapsed time after the timer has been started (by the operator or by a program) and provides an indication when the elapsed time has reached the predefined limit. Timer points can be configured as component or full points.

Timer points are accessible to routines in the same LM, to any LM or PM on the same UCN, to the LC, and to any node on the LCN. A diagram of Logic Manager data paths for a timer point is shown in Figure 3-17, and a diagram of the Timer Data Point processing is shown in Figure 3-18.

To use a Timer data point, an operator at a Universal Station, or a program, loads a preset time value (the length of time that the timer is to run in seconds or minutes) into parameter SP. This value is frozen for one LMM cycle (1/2 second) to assure transfer to the LC. Once the preset value is entered, the timer is started by setting parameter COMMAND to Start. The timer can also be stopped, reset, or reset and restarted through the COMMAND parameter.

The time value in PV starts at zero and increments toward the preset time value. The RV parameter indicates the time remaining until the timer reaches its limit ($SP - PV$). When $PV = SP$, the status parameter S0 is set to ON to indicate that the time limit has been reached. The values in PV and SP can range from 0 to 65535 seconds or minutes, in accordance with the LC program (the parameter TIMEBASE cannot be changed from the Universal Station).

NOTE

The Timer point in an LMM does not itself perform the timer function. The timer function is performed by ladder logic programming in the LC.

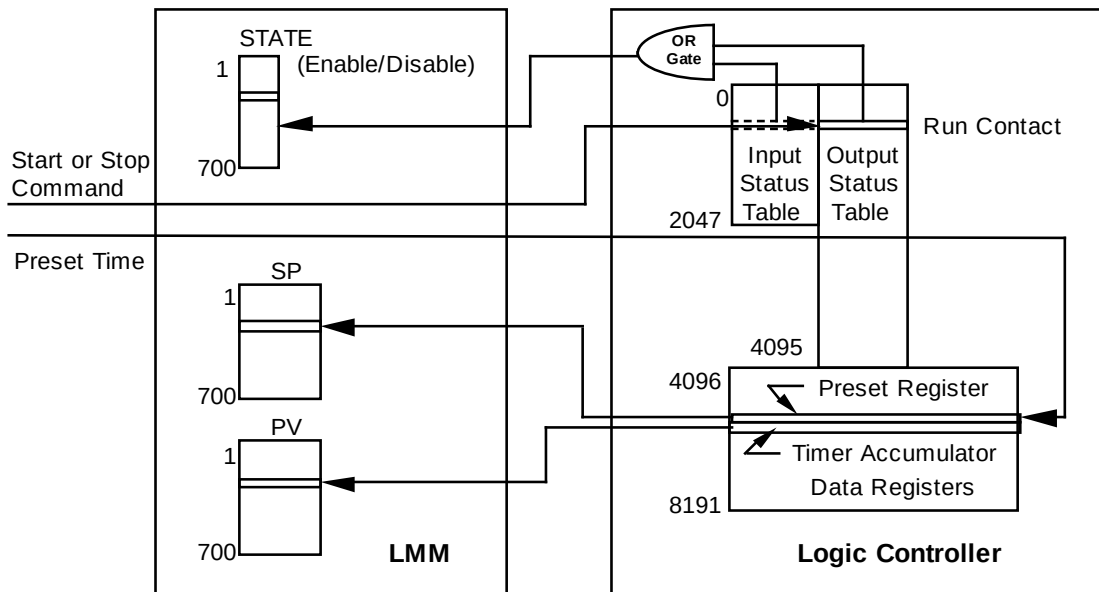


Figure 3-17 — Timer Point Data Paths Between LMM and Logic Controller

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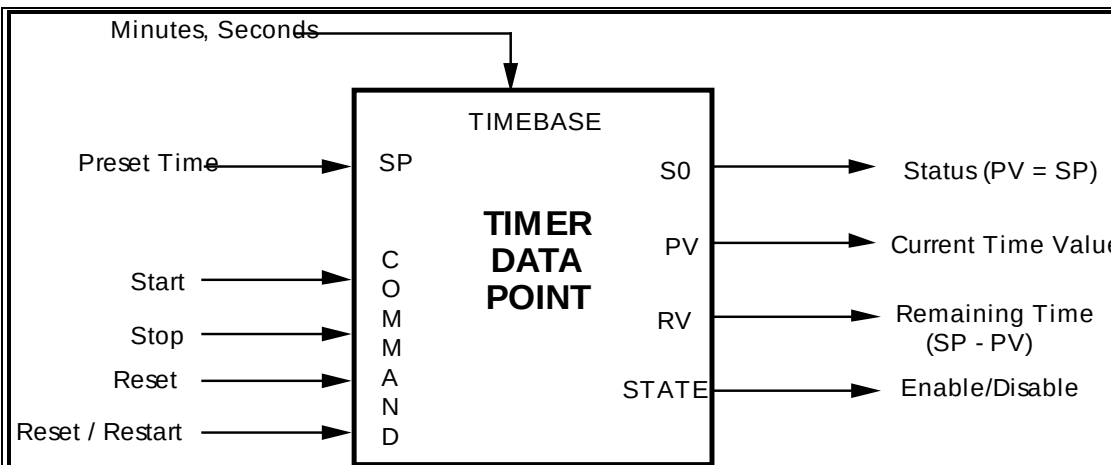


Figure 3-18 — Box Timer Data Point Processing

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Appendix – A

LM I/O AND INDUSTRIAL MICROCOMPUTER MODULES

I/O Modules and Industrial Microcomputer Modules that are available for use with the Logic Manager are tabulated below. For descriptions of these modules, see *620 I/O Specifications User Manual* (Form 620-8995). Note that not all modules listed in the User Manual are valid for the LM.

Table A-1 — Digital Input Modules

Description	No. of Inputs	Model Number
Simulator Input Module	8	621-0009R
115 Vac/dc	8	621-1100R
115 Vac/dc isolated	6	621-1101R
115 Vac	16	621-1160R
115 Vac	32	621-1180R
230 Vac/dc	8	621-1200R
230 Vac/dc isolated	6	621-1201R
24 Vac/dc	8	621-1500R
24 Vac/dc	16	621-1550R
5 Vdc sink	8	621-3300R
12 Vdc sink	16	621-3450R
12-24 Vdc sink	8	621-3500R
12-24 Vdc sink fast response	8	621-3502R
24 Vdc sink	16	621-3550R
24 Vdc sink fast response	16	621-3552R
24 Vdc sink	32	621-3580R

48 Vdc sink	8	621-3600R
48 Vdc sink	16	621-3660R
5 Vdc source	8	621-4300R
5 V TTL	16	621-4350R
12-24 Vdc source	8	621-4500R
12-24 Vdc source fast response	8	621-4502R
24 Vdc source	16	621-4550R

Table A-2 — Digital Output Modules

Description	No. of Outputs	Model Number
Reed Relay	6	621-0007R
115 Vac isolated	8	621-2100R
115 Vac source self-protected	6	621-2101R
115 Vac	8	621-2102R
115 Vac	16	621-2150R
115 Vac	32	621-2175R
230 Vac	8	621-2200R
230 Vac isolated	6	621-2201R
24 Vac	8	621-2500R
24 Vac	16	621-2550R
5 Vdc source	8	621-6300R
5 V TTL	16	621-6350R
12 Vdc source	16	621-6450R
12-24 Vdc source	8	621-6500R
12-24 Vdc source self-protected	8	621-6503R
24 Vdc source	16	621-6550R
24 Vdc low power source	16	621-6551R
24 Vdc source	32	621-6575R

24 Vdc source	8	621-6600R
48 Vdc source	16	621-6650R
120 Vdc source (0.5A)	8	621-6700R
120 Vdc source (2A)	8	621-6701R

Table A-3 — Analog Input Modules

Description	No. of Inputs	Model Number
Thermocouple (mV)	4	621-0014R
Isolated (4-20 mA)	8	621-0022-AR
Isolated (0-10 V)	8	621-0022-VR

Table A-4 — Analog Output Modules

Description	No. of Outputs	Model Number
4-20 mA	4	621-0010-AR
0-10 V	4	621-0010-VR

Table A-5 — Special-Function I/O Modules

Description	Model Number
BCD Converter	621-0006R
Pulse Input	621-0024R
ASCII Communications	621-0012R
Controller Access	621-0016R
Absolute Encoder	621-0018R
High Speed Counter	621-0307R
24 Vdc sink input, source output	621-0576R

Table A-6 — Industrial Microcomputers

Description	Model Number
MiniCOP Module — Double-Wide, 3-Port	627-1002R
MiniCOP Module — Triple-Wide, 4-Port, with RGB Output	627-1003R

Appendix - B

LOGIC MANAGER INSTRUCTION SET

The Logic Manager provides a powerful instruction set, including math functions. A brief definition of the instructions follows. Each instruction and its corresponding opcode and execution time are listed in Table B-1. For more information on the instruction set, consult the MS-DOS Loader User Manual (Form No. 623-8982).

B.1 Relay Logic Instructions

Normally Open Contact	Examines an input for an ON condition; examines an output for an energized condition.
Normally Closed Contact	Examines an input for an OFF condition; examines an output for a de-energized condition.
Transition On Contact	Acts as a one-shot; ON for one scan when its address energizes and OFF thereafter.
Transition Off Contact	Acts as a one-shot; ON for one scan when its address de-energizes and OFF thereafter.
Branch	Creates parallel branch circuits in a logic line.
Output	Energizes when preceding ladder logic is true.
Retentive Output	Logic status is retained ON or OFF when power is removed. At program power-up all retentive outputs assume their last state before the power-down.
Latch Output	Energizes when preceding logic is true and remains energized regardless of logic changes. Must be unlatched to be de-energized.
Unlatch Output	De-energizes a latch output with an identical address when the preceding logic is true. Remains unlatched regardless of the changes in the logic.

B.2 Timer and Counter Instructions

All the timer instructions start from zero and increment toward the preset value.

On Delay Timers (.01, .1, 1 second)

Begin to time when the preceding logic is true. The output turns ON when the accumulated value equals the preset value. Timer is reset to zero when the preceding logic is false.

Off Delay Timers (.01, .1, 1 second)

Begin to time when preceding logic is false. The output turns OFF when the accumulated value equals preset.

Retentive On Delay Timer (.1 and 1 second)

Separate timer RUN and RESET inputs. When the RUN input is false, the timer will not run and the accumulated value is retained. When the RESET input is false, the timer is reset.

Up/Down Counter

Counts from - 65,535 to + 65,535 transferring an ON signal to the output status table and I/O system when the accumulated value equals the preset value.

B.3 Skip Instructions

Not Skip and Retain

When preceding logic is false, all following logic line terminators are skipped, retaining their terminators present status, until a matching End of Skip instruction is encountered.

Not Skip and De-energize

When preceding logic is false, all following logic line terminators are skipped and their terminators de-energized until an End of Skip instruction is encountered.

End of Skip

Marks the point where memory scan terminates skipping and resumes executing lines following Not Skip and Retain or Not Skip and De-energize. When preceding logic is false, all following logic is jumped and not executed until a matching End of Jump instruction is encountered.

Return To Beginning of Program

Instructs program scan to return to the beginning of the program.

B.4 Data Manipulation Instructions

Bring In	Transfers 16 bits of data from the I/O Status Table or 16 bits plus sign bit from the Register Table, to the processor for use within the logic line.
Send Out	Transfers 16 bits of data from the preceding instructions to the I/O system and the Output status or 16 bits plus sign bit to the Register Table.
Pull	Transfers multiple 16-bit groups of data from the I/O modules (with PUSH/PULL capability) or the Register Table, to the LM.
Push	Transfers (with PUSH/PULL capability), or to the Register Table.
Indirect Bring In	Transfers 16 bits of data from the address pointed to by the Indirect Bring In address in the I/O Status or Register Table. Allows multiplexing of input data.
Indirect Send Out	Transfers 16 bits of data from the preceding data instructions, to an address pointed to by the Indirect Send Out address. Allows multiplexing of data.
Constant	Transfers a number between 0 and 65,535 from the user memory to the LM.

B.5 Arithmetic Instructions

Addition	Adds two 16-bit signed numbers.
Subtraction	Subtracts two 16-bit signed numbers.
Multiplication	Multiplies two 16-bit signed numbers.
Division	Divides a 32-bit signed dividend by a 16-bit signed divisor to yield a 16-bit signed quotient and a 16-bit remainder.
Equality Comparison	Compares for equality between two signed data values.
Less Than Comparison	Compares for less-than condition between signed data values.
Greater Than Comparison	Compares for greater-than condition between signed data values.
Test For Zero	Processor tests specify Register or 15 consecutive I/O status table locations for a zero condition. If a zero condition is detected, the contact will be ON.

B.6 Sequencer Instructions

Sequencer	Allows processor to store up to 1,024 16-bit groups of user-defined data in main memory. The data may be used to control repetitive operations or for bulk data storage.
Load Sequencer	Allows data to be transferred to sequencer tables.

WARNING

The Load Sequencer instruction must not be used in redundant systems. It is incompatible with the program download function of the Redundancy Control Module.

Unload Sequencer	Allows data to be transferred from sequencer tables before the sequencer table is executed.
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B.7 Matrix Instructions

Move	Transfers data stored in a specified area of the I/O Status Table or Register Table to another area.
Invert	Inverts and transfers data stored in a specified area of the I/O Status Table or Register Table to another area.
Set Zero	Sets every bit of a matrix to zero.
Set One	Sets every bit of a matrix to one.
Or	Performs an OR of the contents of two matrices and transfers the results to a third matrix.
Exclusive Or	Performs an Exclusive OR of the contents of two matrices and transfers the result to a third matrix.
And	Ands the contents of two matrices and transfers the result to a third matrix.
Compare	Compares two matrices for equality and stores the addresses of mismatches in a third matrix.

B.8 Subroutine Instructions

Subroutine	Allows the processor to execute defined portions of the program on command.
Jump to Subroutine	Causes the processor to jump immediately to an identified subroutine in the program and begin executing the subroutine program.
Return from Subroutine	Identifies the end of a subroutine.

B.9 Miscellaneous Instructions

Input Status Scan	Temporarily suspends program execution in order to allow the Input Status Table to be updated.
No Operation	A place-holding instruction that facilitates on-line program changes.

Table B-1 — Logic Manager Instruction Set, Op Codes, and Execution Times

SYMBOL	INSTRUCTION NAME	HEX CODE	BINARY CODE								EXECUTION TIME	
			2	2	2	2	1	1	1	1	MICRO-SECONDS	
			3	2	1	0	9	8	7	6	MINIMUM	MAXIMUM
--] [--	Normally Open	C8	1	1	D	F	1	0	0	0	1.83	1.83
--]/[--	Normally Closed	CC	1	1	D	F	1	1	0	0	1.83	1.83
--]/\[--	Transition On	43	0	1	D	F	0	0	1	1	9.8	14.8
--]V[--	Transition Off	4B	0	1	D	F	1	0	1	1	9.8	14.8
--NOP--	No Operation	FF	1	1	1	1	1	1	1	1	2.4	2.4
--ISS--	Input Status Scan	8F	1	0	0	0	1	1	1	1	Note A	
N/A	Down Branch	C1	1	1	0	0	0	0	0	1	1.83	1.83
N/A	Double Down Branch	C2	1	1	0	0	0	0	1	0	1.83	1.83
N/A	Up Branch	C3	1	1	0	0	0	0	1	1	1.83	1.83
--(R)--	Retentive Output	C5	1	1	D	F	0	1	0	1	4.27	6.1
--()--	Output Coil	C4	1	1	D	F	0	1	0	0	4.27	6.1
--(L)--	Latch Output	C6	1	1	D	F	0	1	1	0	4.27	6.1
--(U)--	Unlatch Output	C7	1	1	D	F	0	1	1	1	4.27	6.1
--<B2>--	Indirect BringIn	89	1	0	0	0	1	0	0	1	13.51	23.0
--<I2>--	Ind. SendOut (Register)	06	0	0	0	0	0	1	1	0	16.0	86.0 (B)
--<I2>--	Ind. SendOut (Control)	06	0	0	0	0	0	1	1	0	20.0	89.0 (B)
--[B2]--	BringIn	88	1	0	0	0	1	0	0	0	12.7	16.0
--(S2)--	SendOut (Register)	07	0	0	D	F	0	1	1	1	15.0	15.0
--(S2)--	SendOut (Control)	07	0	0	D	F	0	1	1	1	18.0	86.0 (B)
--(PSH)--	Push I/O	84	1	0	0	0	0	1	0	0	Note C	
--(PSH)--	Push Register	86	1	0	0	0	0	1	1	0	Note C	
--[PUL]--	Pull I/O	80	1	0	0	0	0	0	0	0	Note C	
--[PUL]--	Pull Register	82	1	0	0	0	0	0	1	0	Note C	
--[PULS]--	Pull Status	81	1	0	0	0	0	0	0	1	Note C	
--EOS--	End Of Skip	34	0	0	1	1	0	1	0	0	3.0	15.0
--RBP--	Return Beginning Pgm	05	0	0	D	F	0	1	0	1	3.6	5.2
--NSKR--	Not Skip & Retain	46	0	1	D	F	0	1	1	0	21.9	31.72
--NSKD--	Not Skip & De-energize	47	0	1	D	F	0	1	1	1	10.4	24.0
--(TOF)--	TOF Timer .01	40	0	1	1	0	0	0	0	0	43.31	62.83
--(TOF)--	TOF Timer .1	41	0	1	1	0	0	0	0	1	29.28	41.48
--(TOF)--	TOF Timer 1.0	42	0	1	1	0	0	0	1	0	29.28	41.48
--(TOF)--	Word 2		1	1	F	F	0	1	0	0	4.27	6.10
--(TON)--	TON Timer .01	48	0	1	1	0	1	0	0	0	39.04	51.24
--(TON)--	TON Timer .1	49	0	1	1	0	1	0	0	1	27.45	32.23
--(TON)--	TON Timer 1.0	4A	0	1	1	0	1	0	1	0	27.45	37.21
--(TON)--	Word 2		1	1	F	F	0	1	0	0	4.27	6.10
COUNTER	Up/Down Counter	4F	0	1	1	0	1	1	1	1	34.5	51.5
-TIMER	Retentive Timer .1	4D	0	1	1	0	1	1	0	1	23.79	35.38
-TIMER	Retentive Timer 1.0	4E	0	1	1	0	1	1	1	0	23.79	35.38

Table B-1 — Logic Manager Instruction Set, Op Codes, and Execution Times (continued)

SYMBOL	INSTRUCTION NAME	HEX CODE	BINARY CODE								EXECUTION TIME MICRO-SECONDS	
			2 3	2 2	2 1	2 0	1 9	1 8	1 7	1 6	MINIMUM	MAXIMUM
SEQNCR	Sequencer										55.0	109.0
	Sequencer Size	8A	0	1	0	0	1	0	1	0	---	---
	Step Number Address	8C	1	0	0	0	1	1	0	0	---	---
	Data Table Step 1	8B	1	0	0	0	1	0	1	1	---	---
	Control Output Address	07	0	0	0	0	0	1	1	1	15.0	86.0
--(LS2)--	Load Sequencer	44	0	1	0	0	0	1	0	0	3.6	55.0
--(US2)--	Unload Sequencer	45	0	1	0	0	0	1	0	1	3.6	89.0
--SUB--	Subroutine	24	0	0	1	0	0	1	0	0	3.1	11.6
--RTS--	Return to Sub	14	0	0	0	1	0	1	0	0	3.1	37.5
--JSR--	Jump to Sub	04	0	0	0	0	0	1	0	0	19.6	67.0
--]*--	Multiply	0C	0	0	0	0	1	1	0	0	34.43	34.43
--]/--	Divide	0B	0	0	0	0	1	0	1	1	43.45	43.45
--]>--	Greater Than	0A	0	0	0	0	1	0	1	0	14.45	16.45
--]<--	Less Than	09	0	0	0	0	1	0	0	1	115.45	16.45
--]=[--	Equal To	08	0	0	0	0	1	0	0	0	14.45	16.45
--]>Z--	Test For Zero	8D	1	0	0	0	1	1	0	1	11.5	17.25
--]>+--	Add	0E	0	0	0	0	1	1	1	0	25.45	25.45
--]>---	Subtract	0D	0	0	0	0	1	1	0	1	27.45	27.45
MATRIX	Matrix										See individual operator times	
	Size	88	1	0	0	0	1	0	0	0		
	Source Matrix	8E	1	0	0	0	1	1	1	0		
	Reference Matrix	9E	1	0	0	1	1	1	1	0		
	Operator	AE	1	0	1	0	1	1	1	0		
	Destination Matrix	BE	1	0	1	1	1	1	1	0		
	Matrix Instructions Of 8 Real I/O		MATRIX BIT ARRAY 7 6 5 4 3 2 1 0									
	Move	00	0	0	0	0	0	0	0	0	14.0	980.0
	Invert	02	0	0	0	0	0	0	1	0	14.0	1000.0
	Or	04	0	0	0	0	0	1	0	0	14.0	1170.0
	And	06	0	0	0	0	0	1	1	0	14.0	1170.0
	Exclusive Or	08	0	0	0	0	1	0	0	0	14.0	1170.0
	Set 0	0A	0	0	0	0	1	0	1	0	14.0	810.0
	Set 1	0A	0	0	0	0	1	0	1	0	14.0	810.0
	Compare	0C	0	0	0	0	1	1	0	0	14.0	1240.0

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Notes:

- A 1 word = 57.0 + 27.5 per additional word
 1 word = 27.57 + 9.60 per additional word
 1 word = 32.5 + 15.0 per additional word
 1 word = 48.0 + 21.0 per additional word
 1 word = 36.0 + 13.0 per additional word

- C 512 I/O — 71.0 microseconds
 1024 I/O — 1.2 milliseconds
 2048 I/O — 2.06 milliseconds

- B Maximum time for consecutive sendouts. Minimum time is achieved when logic execution time, after send out and before the next line terminator, equals the difference between the minimum and maximum time.

